CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter explains the methodology employed in studying the thermal, hydraulic, and thermodynamic characteristics of the PTSC receiver. The analysis is done numerically in determining the performance impact of internal delta winglet vortex generator and fluid used within the receiver at given operational conditions. As the operation of the system depends on the flow structure, heat transfer process, pressure loss, and fluid properties all these aspects arrangement organized in this methodology.

It starts from describing the physical model, problem formulation, and the numerical approach employed in the current study. After that, the governing equations, the thermophysical properties of the base fluid and hybrid nanofluids are presented as a prerequisite to model the thermal and flow processes occurring inside the receiver tube. Finally, the structure of CFD modeling program which is ANSYS FLUENT and the basic steps of the pre-processing, solving, and post-processing procedures are presented.

Furthermore, the applied thermal boundary conditions, grid independence analysis, and the validation procedure are considered. They constitute an important part of the current study, since they serve the purpose to validate the developed mathematical model. Overall, this chapter describes the main aspects of the methodology utilized in the current study and serves as a basis for discussing the obtained results in the next chapter.

3.2 Physical Model

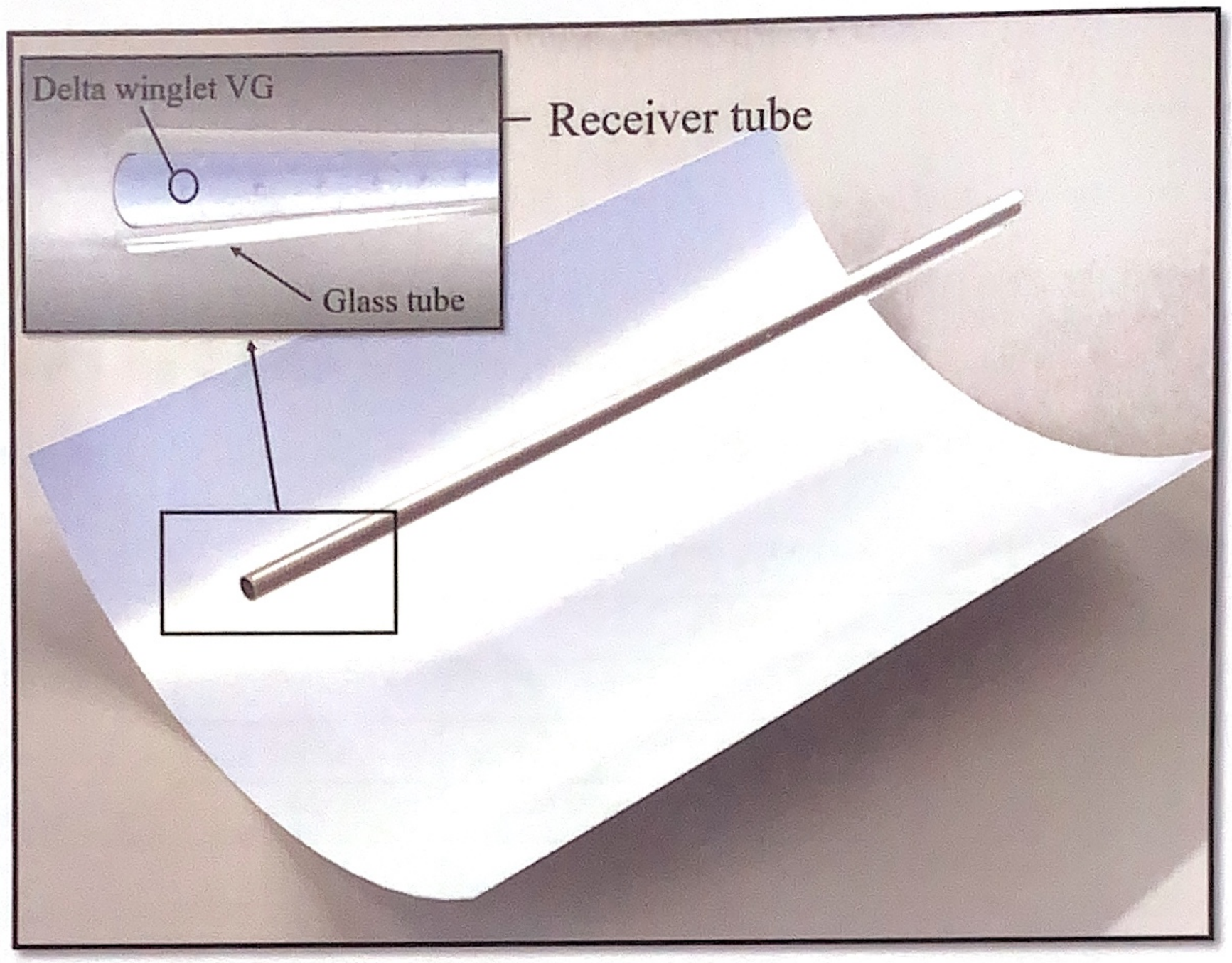
The physical model used in the present work is a parabolic trough solar collector (PTSC) receiver tube shown in Figure 3-1 (a) with a delta winglet insert attached to the side of the tube. The receiver consists of outer borosilicate glass envelope and inner steel absorber tube with a cermet coating, and the flowing heat transfer fluid inside the absorber tube with dimensions shown in Table 3-1. The absorber tube and the vortex generators are made of steel and aluminum, respectively. The thermal conductivity values for the steel absorber tube and the aluminum vortex generators used in the numerical analysis were 16 W/m·K and 202.4 W/m·K, respectively.

Within the absorber tube, the delta-winglet insert serves as a passive heat transfer enhancement structure. The design parameters of the insert as shown in Figure 3-1 (b) and (c) are the attack angle α, blockage ratio BR (B/D), and pitch ratio PR (P/D), defining the alignment, flow blockage, and spacing between vortex generators, respectively. These design parameters determine the intensity of secondary flow generation and the level of interference with thermal boundary layer development inside the absorber tube.

In addition to the base fluid (Syltherm-800), four hybrid nanofluids including Al_2O_3-GO/Syltherm, Fe_2O_3-GO/Syltherm, Al_2O_3-SiO_2/Syltherm, and Cu-MgO/Syltherm were used. With the use of the geometrical configuration of the PTSC receiver, delta-winglet insert, and the chosen hybrid nanofluids under one computational model, the physical model forms the foundation for the investigation of the overall thermal, hydraulic, and thermodynamic performance of the receiver.

Table 3-1: Geometrical dimensions of the PTSC receiver tube.

Dimension	Value (mm)
Absorber tube length	3000
Absorber inner tube inside diameter (d_{ri})	66
Absorber inner tube outside diameter (d_{ro})	70
Glass outer tube inside diameter (d_{gi})	109
Glass outer tube outside diameter (d_{go})	115



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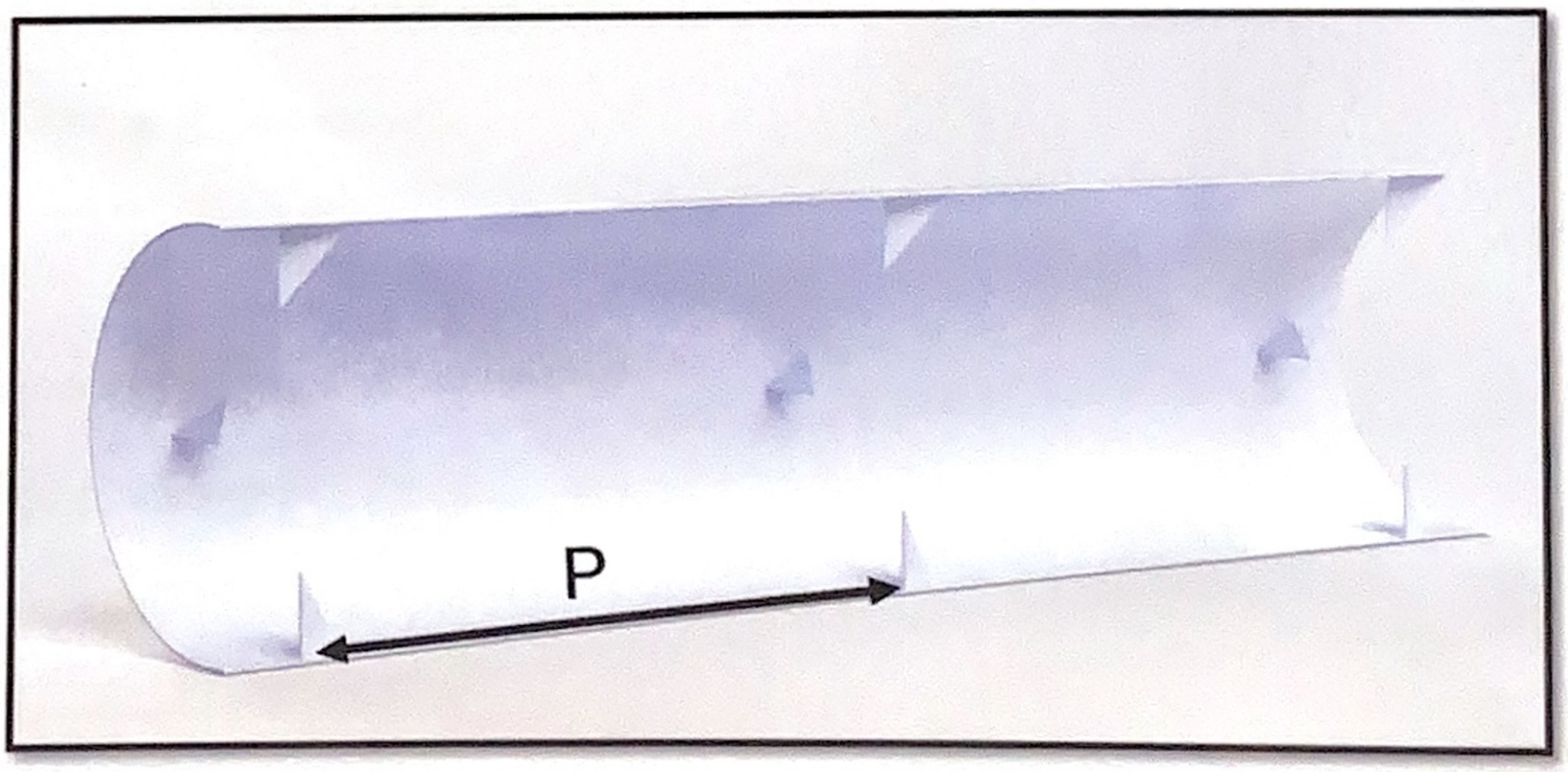

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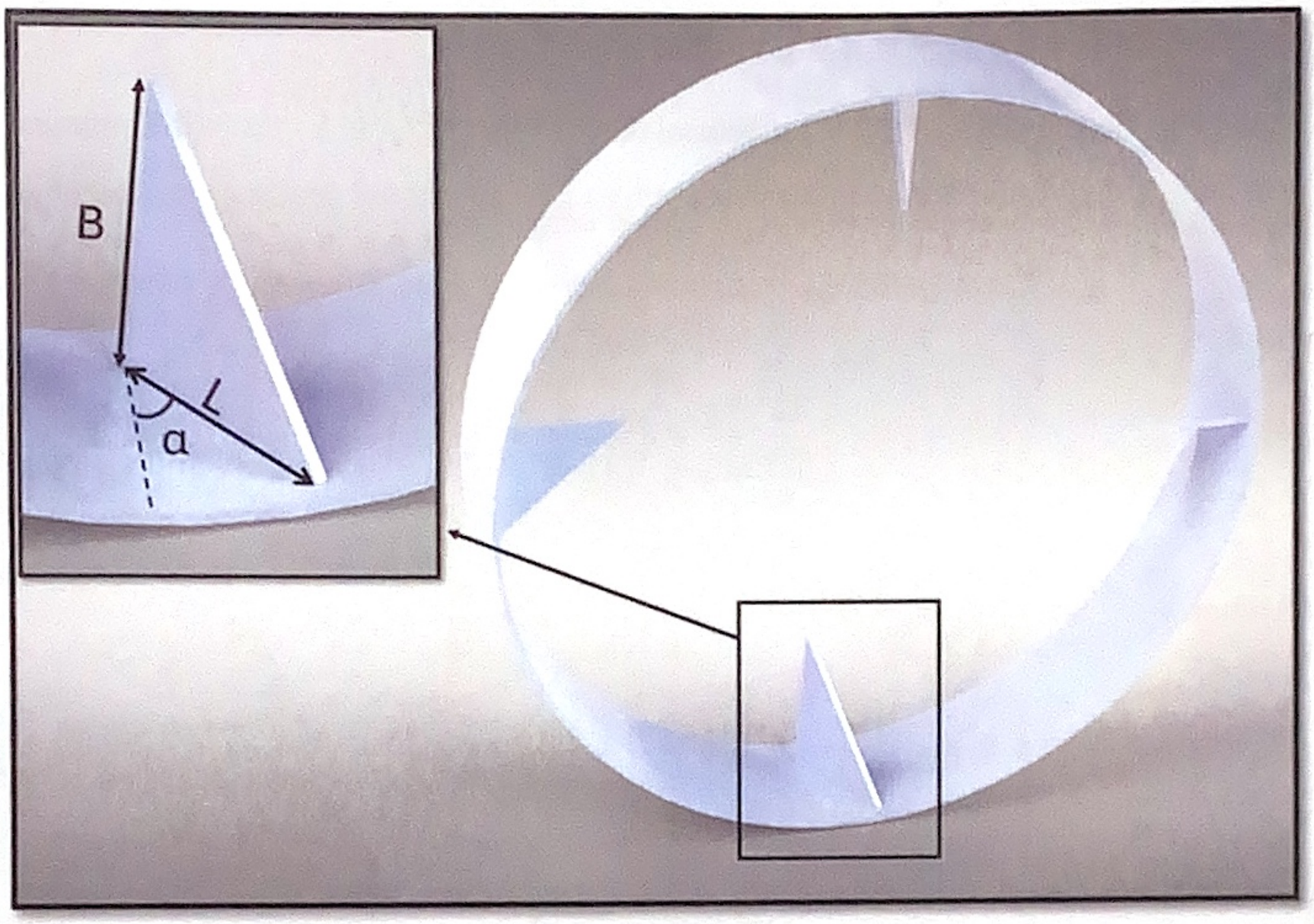


(c)

Figure 3-1 Geometrical model of the PTSC receiver with internal delta-winglet insert

3.3 Governing Equations

The Navier-Stocks equations (NSE) that control the flow and thermal fields were employed in modeling the PTSC's absorber tube as indicated below[66].

Continuity:

$$\frac{\partial(\rho\bar{u}_i)}{\partial x_i} = 0 \tag{3.1}$$

Momentum:

$$\frac{\partial}{\partial x_i}(\rho\bar{u}_i\bar{u}_j) = -\frac{\partial\bar{P}}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial\bar{u}_i}{\partial x_j} + \frac{\partial\bar{u}_j}{\partial x_i}\right) - \frac{2}{3}\mu\frac{\partial\bar{u}_i}{\partial x_i}\delta_{ij} - \left(\rho\overline{u_i'u_j'}\right)\right] \tag{3.2}$$

Energy:

$$\begin{aligned}\frac{\partial}{\partial x_j}(\rho\bar{u}_j C_p\bar{T}) =& \frac{\partial}{\partial x_j}\left(k\frac{\partial\bar{T}}{\partial x_j} + \frac{\mu_t}{\sigma_{ht}}\frac{\partial(C_p\bar{T})}{\partial x_j}\right) + \bar{u}_j\frac{\partial\bar{P}}{\partial x_j} \\ &+ \left[\mu\left(\frac{\partial\bar{u}_i}{\partial x_j} + \frac{\partial\bar{u}_j}{\partial x_i}\right) - \frac{2}{3}\mu\frac{\partial\bar{u}_i}{\partial x_i}\delta_{ij} - \left(\rho\overline{u_i'u_j'}\right)\right]\frac{\partial\bar{u}_i}{\partial x_j}\end{aligned} \tag{3.3}$$

To analyze the Reynolds stresses ($-\rho\overline{u_i'u_j'}$) in Eq. (3.2), k-ε turbulence model was selected for this purpose. In Eqs. (3.2) and (3.3), $\bar{u}_i$ and $\bar{u}_j$ represented the time-averaged velocities in the i-direction and j-direction, respectively. $\bar{T}$ and $\bar{P}$ are the time-averaged temperature and pressure, respectively. To compute the Reynolds stresses components of the RANS equations, the eddy viscosity approach of the Boussinesq approximation assumption that linked the stresses with the strain rate of the form [66]:

$$\left(-\rho\overline{u_i'u_j'}\right) = \mu_t\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - \frac{2}{3}\left(\rho k + \mu_t\frac{\partial u_k}{\partial x_k} + \frac{\partial\bar{u}_j}{\partial x_i}\right)\delta_{ij} \tag{3.4}$$

where ***k*** is the turbulent kinetic energy per unit mass and it is given by:

$$\boldsymbol{k} = \frac{1}{2}\left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}\right) \tag{3.5}$$

The transport equations for the turbulent kinetic energy ($\boldsymbol{k}$) and the turbulent dissipation rate (ϵ) used in the realisable k-epsilon turbulence model are given by:

$$\frac{\partial}{\partial x_j}(\rho \boldsymbol{k} \bar{u}_j) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial \boldsymbol{k}}{\partial x_j}\right] + G_k - \rho\varepsilon \tag{3.6}$$

$$\frac{\partial}{\partial x_j}(\rho \varepsilon \bar{u}_j) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + \rho C_1 S_\varepsilon - \rho C_2 \frac{\varepsilon^2}{\boldsymbol{k} + \sqrt{v_\varepsilon}} \tag{3.7}$$

Where G_k is the production of turbulent kinetic energy given by:

$$G_k = -\rho\overline{u_i u_j}\frac{\partial u_j}{\partial x_i} \tag{3.8}$$

energy (G_k) can be obtained as:

$$G_k = \mu_t S^2 \tag{3.9}$$

The turbulence viscosity is calculated by:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \tag{3.10}$$

Detailed determination of C_μ is given in Ref. [66]. The model constants for the realizable k-epsilon turbulence model are:

$C_1 = \max\left[0.43, \frac{\eta}{\eta+5}\right]$, $\eta = S\frac{k}{\epsilon}, S = \sqrt{2S_{ij}S_{ij}}$, $C_2 = 1.9, \sigma_k = 1.9, \sigma_\varepsilon = 1.2$, and S_{ij} represents the rate of linear deformation of a fluid element.

The discrete ordinates radiation model equation is [66]:

$$\frac{dI(\vec{r},\vec{s})}{ds} + (a+\sigma_s)I(\vec{r},\vec{s}) = an^2\frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi}\int_0^{4\pi} I(\vec{r},\vec{s}')\Phi(\vec{r},\vec{s}')d\Omega' \qquad (3.11)$$

where $\vec{r}$ is the position vector, $\vec{s}$ is the direction vector, $\vec{s}'$ is the scattering direction vector, s is the path length, a is the absorption coefficient, n is the refractive index, σ is the Stefan-Boltzmann constant, σ_s is the scattering coefficient, I is the radiation intensity, which depends on position $\vec{r}$ and direction $\vec{s}$, T is the local temperature, Φ is the phase function, and Ω' is the solid angle.

3.4 CFD Code Structure

Workflow organized into three parts pre-processing, solver, and post-processing modules [67].

3.4.1 Pre-processing

The three-dimensional receiver geometry variants were built using Design Modeler, variants include the smooth receiver and altered shape with delta winglet inserts. Generated meshes were fitted to the geometric surface with tetrahedrons dominant elements. Local refinement of meshes near the wall grid points implemented to improve the modeling of the boundary layer of the fluid. Mesh quality was checked until the constraint conditions are met. Materials (syltherm-800 and hybrid nanofluids) thermophysical properties were considered.

3.4.2 Solver

All the equations have been solved in ANSYS FLUENT with pressure-based segregated flow solver using finite volume method. All simulations were carried out using steady state solution assumption and the working fluid was assumed to be incompressible and was modeled as single phase flow. Moreover, the simulation was done for a non-uniform heat flux on the outer surface of the absorber tube, which was a realistic representation of thermal loading in the PTSC receiver. All the convective terms in the governing transport equations, including the realizable k-ε turbulent model and Discrete Ordinates (DO) radiative transfer equation were discretized by the second-order upwind scheme, while all diffusion terms were discretized using central differencing. Coupled algorithm was used for pressure-velocity coupling. Finally, enhanced wall treatment with low y+ was used to enhance predictions of near wall flows and heat transfer behaviors. The convergence criteria used in this study are 10^{-8} and for the energy equation, 10^{-6} for DO model, and 10^{-5} for all other variables.

3.4.3 Post-processing

Data collected from solution included plots, contours, particle tracks, surface or line averages, and data export for further analysis. Data computed from converged solutions:

friction factor calculated as [68]:

$$\Delta P = f \frac{L}{d_{ri}} \frac{\rho u_m^2}{2} \tag{3.12}$$

Where f, ρ, d_{ri}, u_m, and L are the friction factor, hybrid nanofluid density, the absorber tube's inner diameter, the mean velocity in the absorber tube and the length of the tube, respectively.

The fanning friction factor is evaluated using the following equation:

$$f = \frac{2\tau_w}{\rho u_m^2} \tag{3.13}$$

where τ_w is the shear stress.

The heat transfer coefficient (h) and Nusselt number (Nu) are calculated as follows [68]:

$$h = \frac{q'}{(T_w - T_b)} \tag{3.14}$$

where T_w, and T_b are the average temperature of the absorber tube's inner wall and the bulk fluid temperature, respectively.

$$Nu = \frac{hd_{ri}}{\mathrm{k}} \tag{3.15}$$

where k and h are the thermal conductivity and average heat transfer coefficient, respectively.

The flow Reynolds number is calculated as [68]:

$$\mathrm{Re} = \frac{\rho u_m d_{ri}}{\mu} \tag{3.16}$$

where u_m is the mean velocity and d_{ni} is the absorber tube's inner diameter.

The circumferential temperature gradient of the absorber tube is defined as [68]:

$$\Delta T = T_{abs.\max} - T_{abs.\min} \tag{3.17}$$

Thermal efficiency [68]:

$$\eta_{th} = \frac{\dot{Q}_u - P_p/\eta_{el}}{\dot{Q}_{solar}} \tag{3.18}$$

$$Q_u = \dot{m}\, c_p (T_{out} - T_{in}) \tag{3.19}$$

$$\dot{Q}_{solar} = A_a I_b \tag{3.20}$$

In the above equation, $\dot{Q}_u = \dot{m} C_p (T_{\text{out}} - T_{\text{in}})$ is the heat transfer rate, A_a is the collector's aperture area, I_b is the incident solar radiation, $P_p = \dot{V}\Delta P$ is the pumping power and η_{el} is the product of typical efficiencies of the power block, the electrical generator, and the heat transfer fluid pump and it was assumed 32.7%.

The performance evaluation criterion factor (PEC) is calculated based on the fluid used (PEC) and based on the passive technique used (PEC_{h}) as given by [69]:

$$PEC = \left(\frac{Nu_D}{Nu_o}\right) / \left(\frac{f_D}{f_o}\right)^{1/3} \tag{3.21}$$

$$PEC_{\text{h}} = \left(\frac{h_{hnf}}{hu_f}\right) / \left(\frac{f_{hnf}}{f_f}\right)^{1/3} \tag{3.22}$$

Where the subscripts hnf, f, D and o refer to hybrid nanofluids, base fluid (Sylltherm oil), PTSC with delta-winglets, and PTSC with smooth tube respectively.

The total entropy generation comes from irreversibility of heat transfer (S^H_{gen}) and fluid friction (S^F_{gen}) and its formula is given:

$$\left(S^T_{\text{gen}}\right) = \left(S^F_{\text{gen}}\right) + \left(S^H_{\text{gen}}\right) \tag{3.23}$$

The entropy generation caused by irreversibility of fluid friction is calculated by:

$$\left(S^F_{\text{gen}}\right) = \frac{\mu}{T}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\frac{\partial u_i}{\partial x_j} + \frac{\rho\varepsilon}{T} \tag{3.24}$$

The entropy generation due to irreversibility of heat transfer is calculated by:

$$\left(S_{\text{gen}}^{H}\right) = \frac{k}{T^2}(\nabla T)^2 + \frac{\alpha_t}{\alpha}\frac{k}{T^2}(\nabla T)^2 \tag{3.25}$$

Where α and α_t are the thermal diffusivities, and k is the fluid's thermal conductivity. The first part of the equation represents entropy generation from mean temperature heat transfer while the second part represents entropy generation from fluctuating one.

The total entropy generation rates are first calculated for each control volume, then the total entropy generation rate over the whole computational domain was then calculated by integrating them over the entire volume V by:

$$\left(S_{\text{gen}}^{T}\right) = \int_{V} S_{\text{gen}}\, dV \tag{3.26}$$

The Bejan number, presents the irreversibility caused by heat transfer to the total entropy generation rate, is described as [70]:

$$Be = \frac{\left(S_{\text{gen}}^{H}\right)}{\left(S_{\text{gen}}^{T}\right)} \tag{3.27}$$

The irreversibility of heat transfer is predominant when $\text{Be} = 1$, and the irreversibility of fluid friction is prevailing when $\text{Be} = 0$.

The entropy generation ratio (Ns), presents the total entropy generation of PTSC having delta-winglets to the total entropy generation of smooth PTSC, is described as:

$$Ns = \frac{\left(S^{T}_{\text{gen}}\right)_{D}}{\left(S^{T}_{\text{gen}}\right)_{0}} \tag{3.28}$$

The exergetic efficiency is described as the ratio of the useful work output of the system to the reversible work output for work consuming systems and it can be assessed by Ref. [38]:

$$\eta_{\text{exr}} = \frac{Ex_u}{Ex_a} = \frac{\dot{m}C_{pf}\left[\left(T_{\text{out}} - T_{\text{in}}\right) - T_{\text{amb}} \ln\left(\frac{T_{\text{out}}}{T_{\text{in}}}\right)\right]}{A_a I_b \left[1 + \frac{1}{3}\left(\frac{T_{\text{amb}}}{T_s}\right)^4 - \frac{4}{3}\left(\frac{T_{\text{amb}}}{T_s}\right)\right]} \tag{3.29}$$

Where T_{amb} is the ambient temperature, which is considered 300 K in this work, and T_s is the sun's surface temperature, which is considered 5800 K.

3.5 Optical Model and Thermal Boundary Conditions

SolTrace is first used to determine the concentrated solar heat flux distribution on the receiver tube as shown in Figure 3-2 which presents rays hitting the receiver tube. SolTrace is used for ray trancing based on LS-2 PTSC model to calculate local concentration ratio (LCR). The results of the LCR profile are presented in Figure 3-3 (a). This heat flux map can then be coupled with ANSYS Fluent using a User Defined Function (UDF) shown in APPENDIX A to set up the thermal boundary condition on the wall surface. By doing so, simulations can be done using certain simplifications and boundary conditions to study the thermal behavior of the receiver numerically. The temperature distribution on the receiver tube wall shown in Figure 3-3 (b).

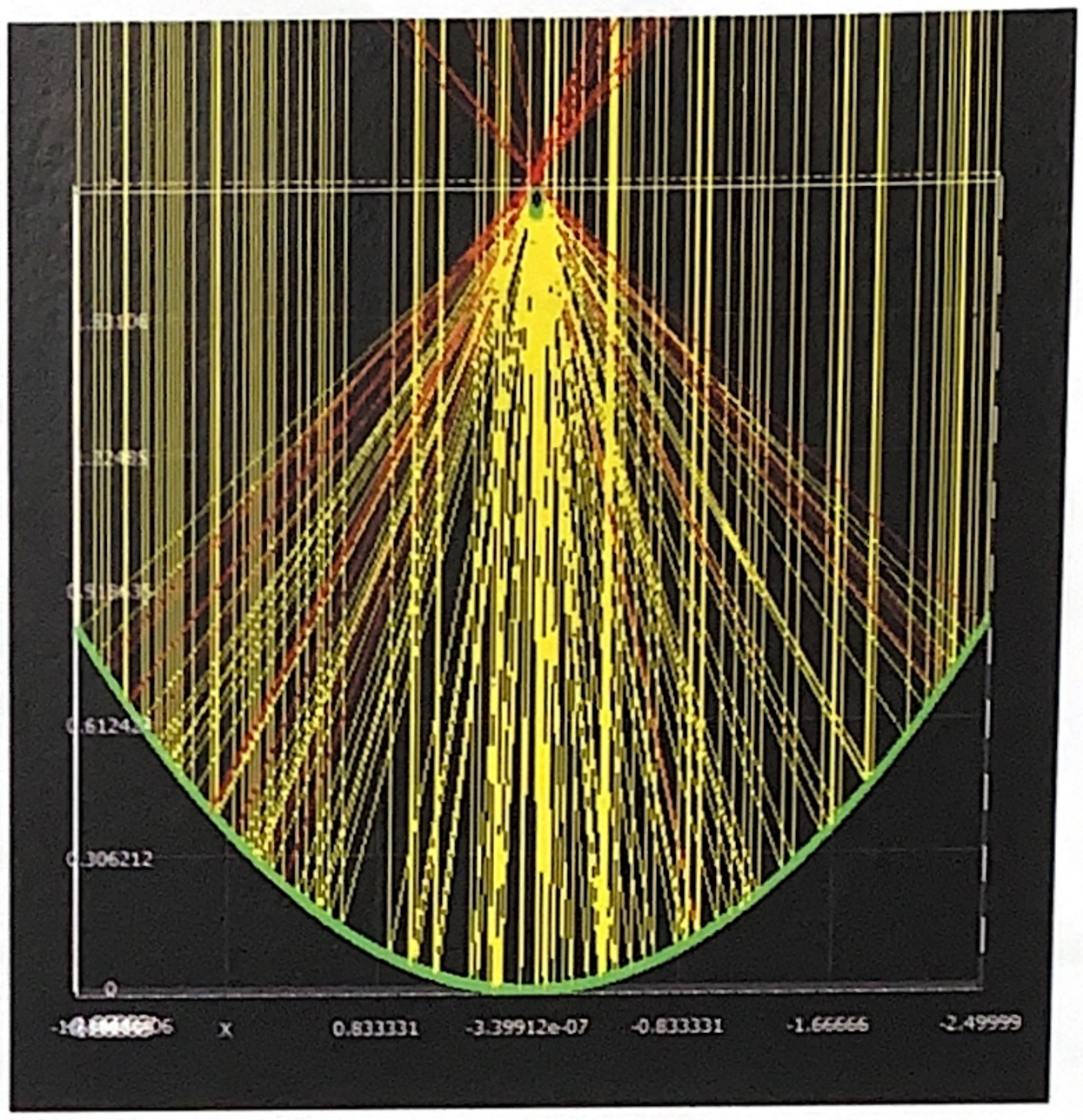
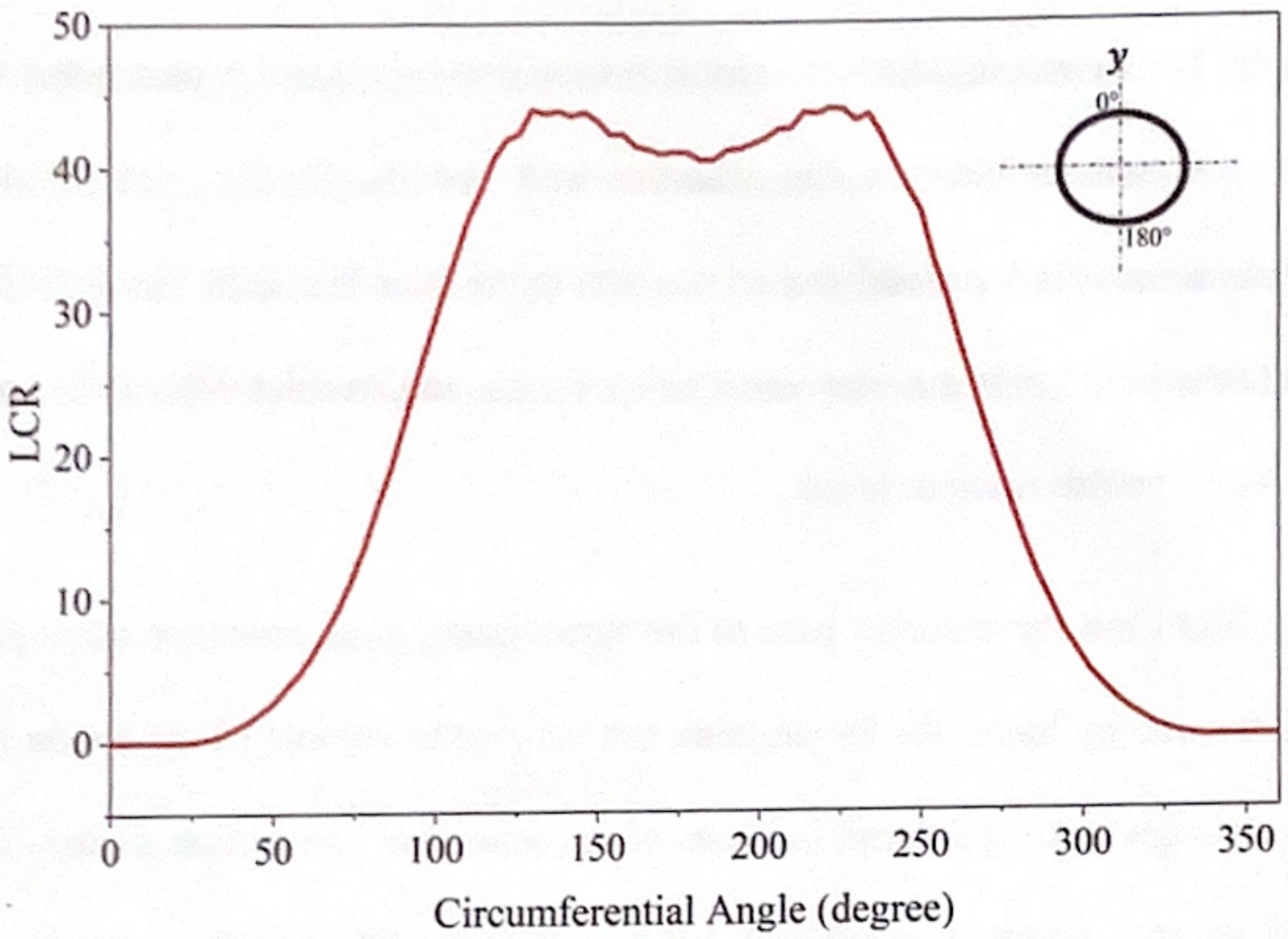


Figure 3-2: Solar-ray intersections obtained from SolTrace for the PTSC geometry.



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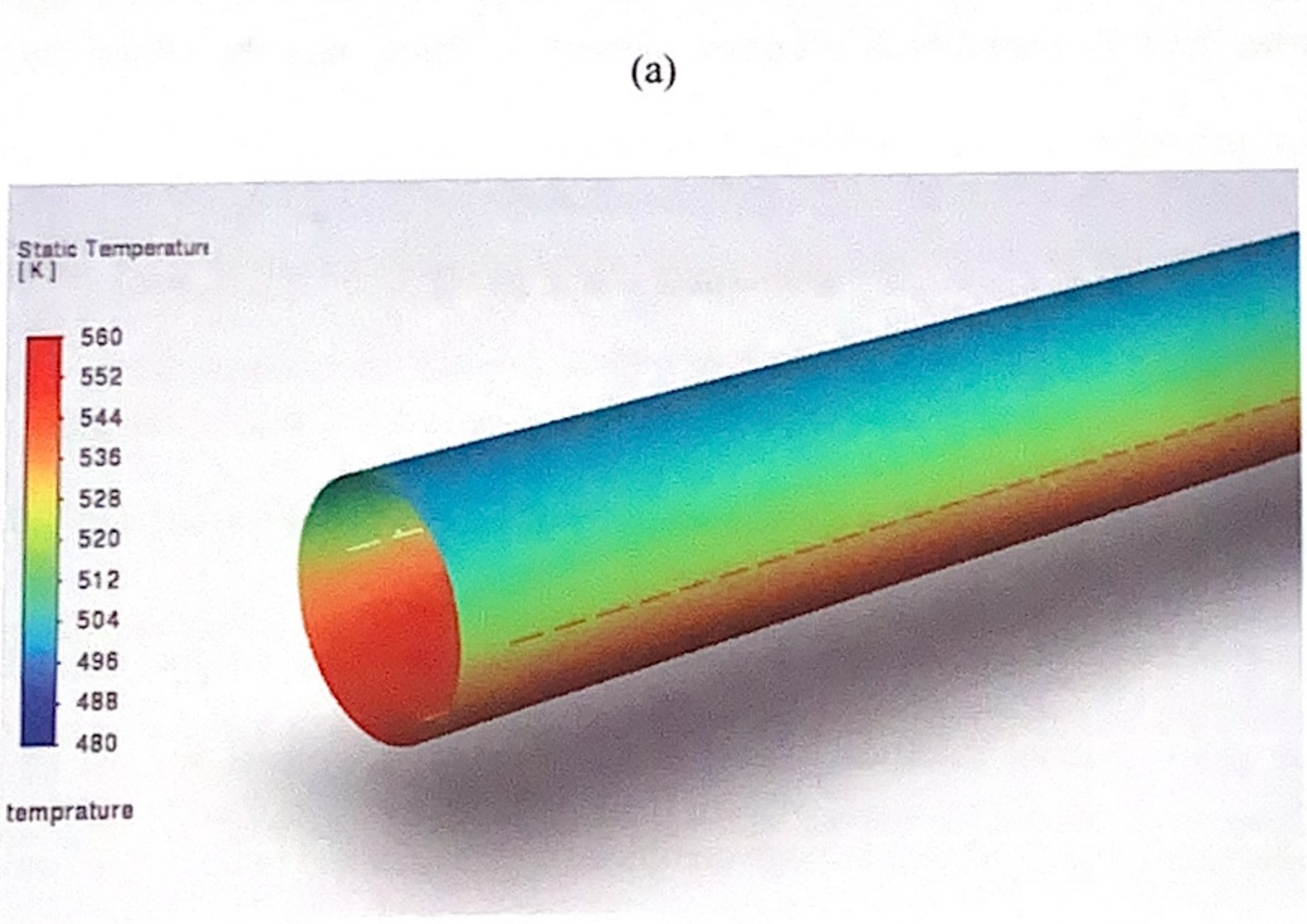


(b)

Figure 3-3: Non-uniform solar-flux distribution around the absorber tube: (a) variation of local concentration ratio (LCR) with circumferential angle, and (b) resulting wall-temperature distribution

The majority of the heat transferred to the outside wall of the absorber tube is conducted to the inside wall and absorbed by the fluid passing through it by the process of forced convection. In this investigation, the absorber tube is assumed to be constructed from steel, and the heat transfer between the absorber wall and the fluid is simulated by using conduction across the tube wall and convection to the fluid inside it. The flow fluid in the absorber tube is fed with a certain inlet temperature, and its inlet velocity is specified by the chosen Reynolds number range.

The heat loss from the absorber tube to the surrounding glass envelope takes place in the annulus formed by them. As the annulus has very little volume of air inside it, the heat transfer through it is considered in terms of radiation and convection modes. In order to calculate the radiative interaction within the annulus, the DO radiation model is used. The absorber tube is coated with a cermet selective coating, and the emissivity value is considered temperature-dependent and is obtained from [71]:

$$\varepsilon = 0.062 + 2 \times 10^{-7} T^2 \tag{3.30}$$

Whereas T is the absorber tube temperature in K. The inner surface of the glass enclosure is assumed to be a gray body with an emissivity of one.

The heat absorbed at the inner surface of the glass enclosure will conduct through the wall of the glass material thickness and finally leave from the outer surface of the glass enclosure to the environment through combined convection and radiation. Glass enclosure will be made up of borosilicate glass material with thermal conductivity of 1.15 W/m.K. Therefore, the outer surface of the glass cover experiences a combined convection boundary condition with the ambient environment and radiation with the sky. Radiative

heat transfer is calculated using Stefan-Boltzmann equation, whereas external convection heat transfer coefficient, sky temperature, and sky emissivity values are obtained from empirical correlations. Ambient conditions and wind velocities for the current study will be based on PTSC operating condition. The external emissivity of the external surface of the glass envelope is assumed to be 0.86. The convection heat transfer coefficient, sky temperature and emissivity for heat transfer by radiation are calculated by the following correlations [72], [73]:

$$h_w = 4v_w^{0.58} d_{go}^{-0.42} \tag{3.31}$$

$$T_{sky} = 0.0552 T_{amb}^{1.5} \tag{3.32}$$

$$\varepsilon_{sky} = 0.711 + 0.56\left(\frac{T_{dp} - 273.15}{100}\right) + 0.73\left(\frac{T_{dp} - 273.15}{100}\right)^2 \tag{3.33}$$

where v_w is the wind velocity and assumed to be 2 m/s in this study, d_{g_0} is the glass cover outer diameter, T_{amb} is the surrounding temperature (assumed 300 K in this study), and T_{dp} is the dew point temperature (assumed to be 293 K).

3.6 Grid-Independence Study

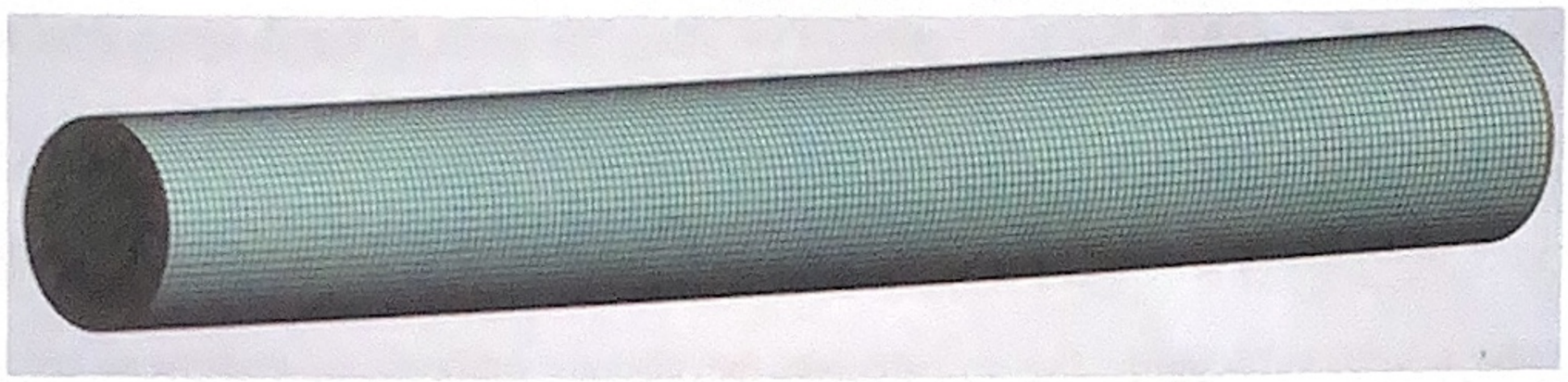
In order to provide justification for the accuracy of the numerical solutions, a detailed calculation procedure had to be performed to determine the total number of meshes that provide appropriate array solutions to determine the flux and temperature distribution. The mesh topology obtained for the entire geometry for the PTSC. It can be observed that for the CFD calculation of the fluid region, tetrahedral meshing is performed along with a highly refined grid system for the region closer to the PTSC's inner wall. Whereas, for the

PTSC tube's outer wall, air, and glass tube, a hexahedral mesh is performed as shown in Figure 3-4. The evolution of average Nusselt numbers for the Reynolds number of 20000 along with the inlet temperature of 500K is provided in Table 3-2. It can be noted from these calculations that the grid system of model 4, totaling 5,692,550 elements as shown in Table 3-2, is quite dense, and hence this study adopts this grid system, since the maximum difference in Nu number was found to be less than 1%, implying that the result is independent of the grid number, adopting the larger increases solution time.

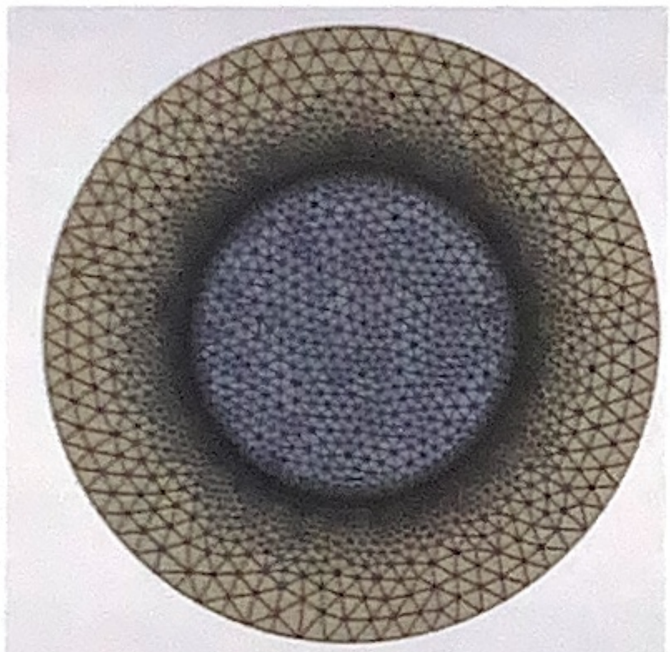
Table 3-2: Grid-independence study results

Grid	Elements	Nu	f	ΔNu (%)	Δf (%)
1	4,848,333	174.2209	0.100732	—	—
2	5,135,296	164.8318	0.102992	-5.39%	2.24%
3	5,398,975	161.2826	0.105990	-2.15%	2.91%
4	5,692,550	160.2266	0.107970	-0.65%	1.87%
5	6,123,283	159.8787	0.110051	-0.22%	1.93%
6	6,363,427	158.5882	0.109677	-0.81%	-0.34%

The largest corrections occur between the coarser grids, while the solution becomes progressively more stable at higher element counts. Beyond roughly 5.7 million cells, the changes in Nu and f become comparatively modest, which indicates that the mesh is sufficiently refined for the comparative parametric analysis.



(a)



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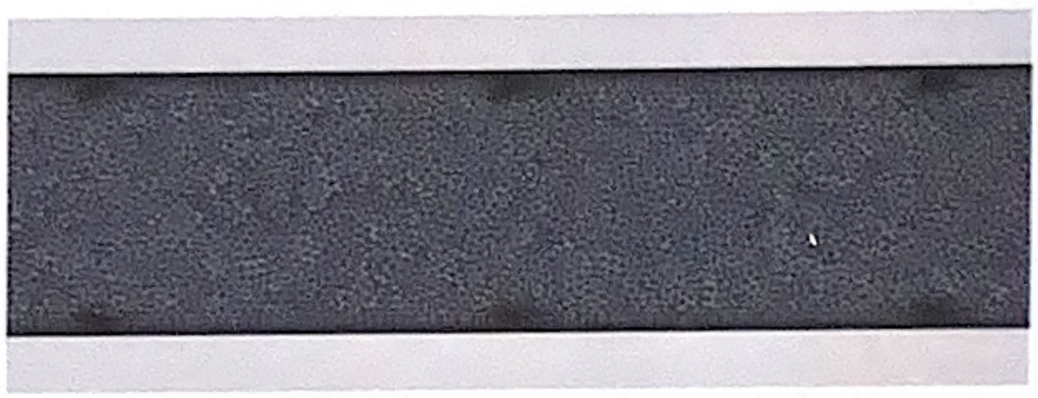

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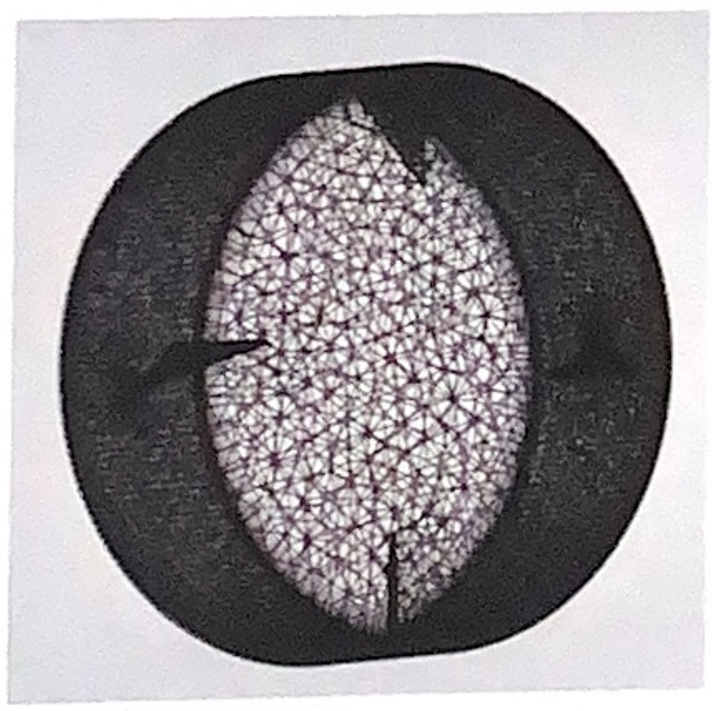

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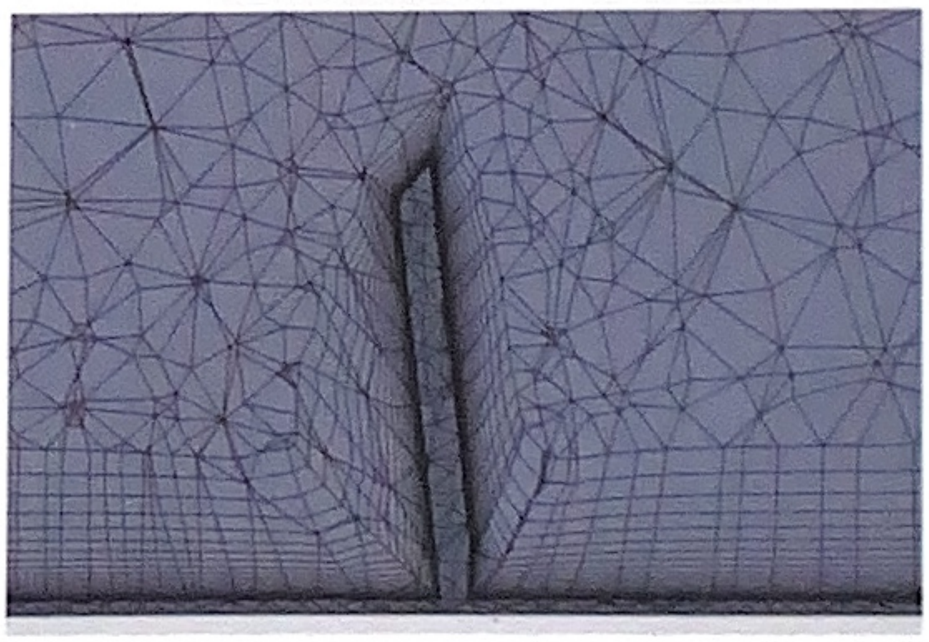


(e)

Figure 3-4: Computational mesh topology of the PTSC receiver with internal delta-winglet vortex generators: (a) full receiver mesh, (b) front mesh of the absorber tube, (c) longitudinal sectional mesh of the receiver, (d) 3-D transparent mesh showing the delta-winglet arrangement, and (e) local mesh refinement around the vortex generator and near-wall region..

As shown in Figure 3-5, the distribution of y+ along the walls of the absorber tube and delta-winglet insert is plotted for the chosen mesh grid. It can be seen that the values of y+ are very small along the wall throughout, generally less than 0.3 in most cases. It implies that the grid is sufficiently fine to calculate the viscous sublayer, as confirmed by the Enhanced Wall Treatment adopted in the numerical simulation. On the other hand, the contour plot reveals that there is an appearance of somewhat larger values of y+ around the local areas in the vicinity of the winglet tips and junctions where strong flow acceleration and shear take place. However, they lie in a very small range, indicating that the first near-wall cell location is correct and that the boundary layer region is successfully captured.

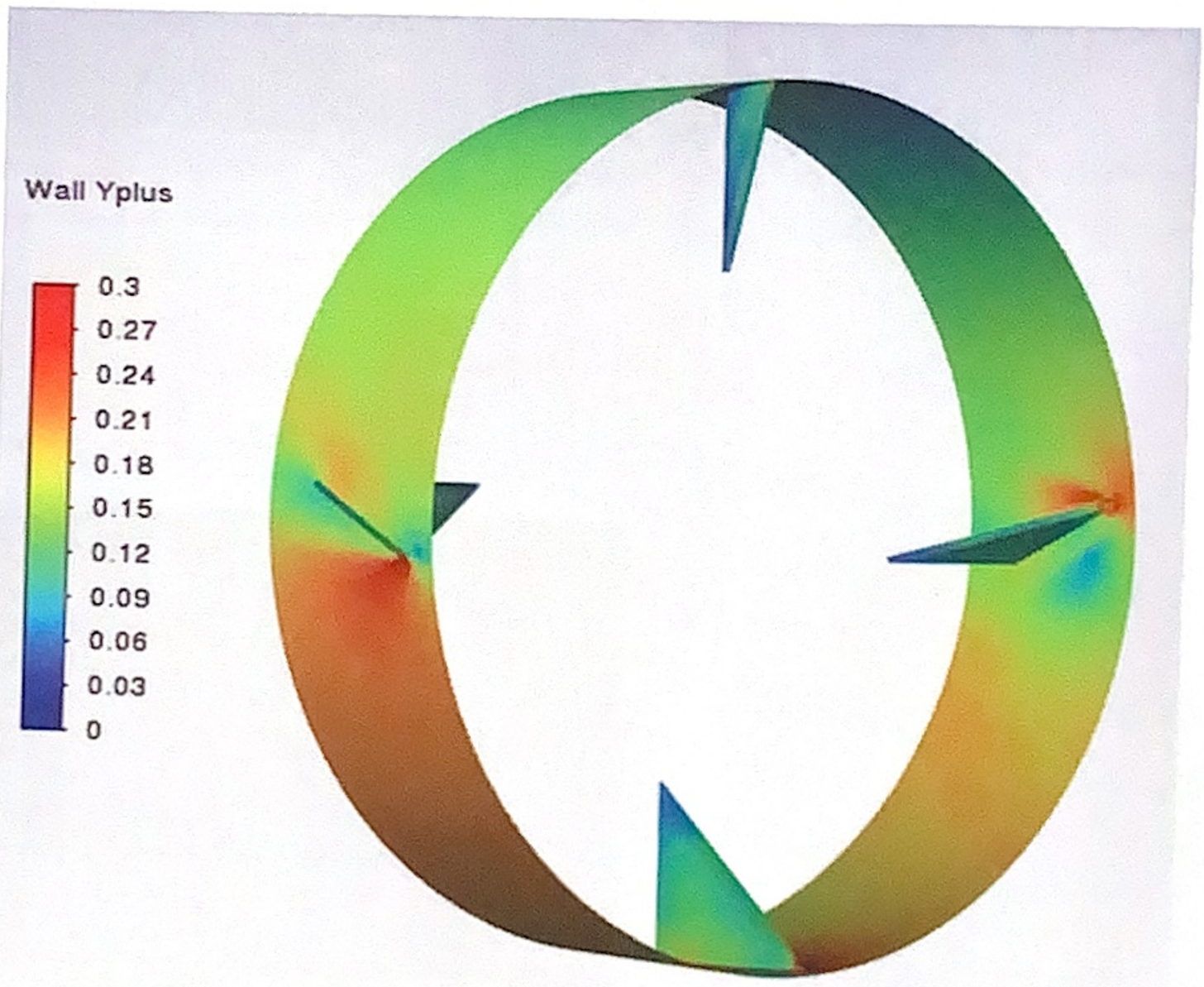


Figure 3-5: Dimensionless wall distance (y+) distribution on the vortex generator surface

3.7 Model Validation

Before exploring new geometries or fluids, the numerical setup is validated against trusted models for smooth PTSC receivers. Comparisons include average Nusselt number, and friction factor reported in prior experimental and numerical studies [68]. Agreement within typical CFD uncertainty bounds, together with strict conservation checks and mesh independence, is taken as evidence that the model can be used for the parametric studies that follow.

3.7.1 Empirical Correlations

For the validation of the proposed numerical model, the Nusselt number and the friction factor corresponding to the smooth-tube configuration have been tested with the aid of the existing correlation equations in the literature. Heat transfer and friction factor results were verified against Gnielinski and Petukhov's correlations, respectively. As illustrated in Figure 3-6, there is an excellent agreement between the present CFD results and the existing experimental correlations. CFD results follow the same trend as the correlation equations, showing very little deviation.

Gnielinski Correlation [74] (valid for $2300 < \mathrm{Re} < 5 \times 10^6, 0.5 < \mathrm{Pr} < 2000$):

$$Nu = \frac{(f/8)(Re-1000)Pr}{1+12.7(f/8)^{1/2}(Pr^{2/3}-1)} \tag{3.34}$$

Petukhov Correlation [75] (valid for $10^4 < Re < 5 \times 10^6$):

$$Nu = \frac{(f/8)\mathrm{Re}Pr}{1.07+12.7(f/8)^{1/2}(\mathrm{Pr}^{2/3}-1)} \tag{3.35}$$

Petukhov's relation for friction factor [75]

$$f = (0.79\ln(Re) - 1.64)^{-2} \tag{3.36}$$

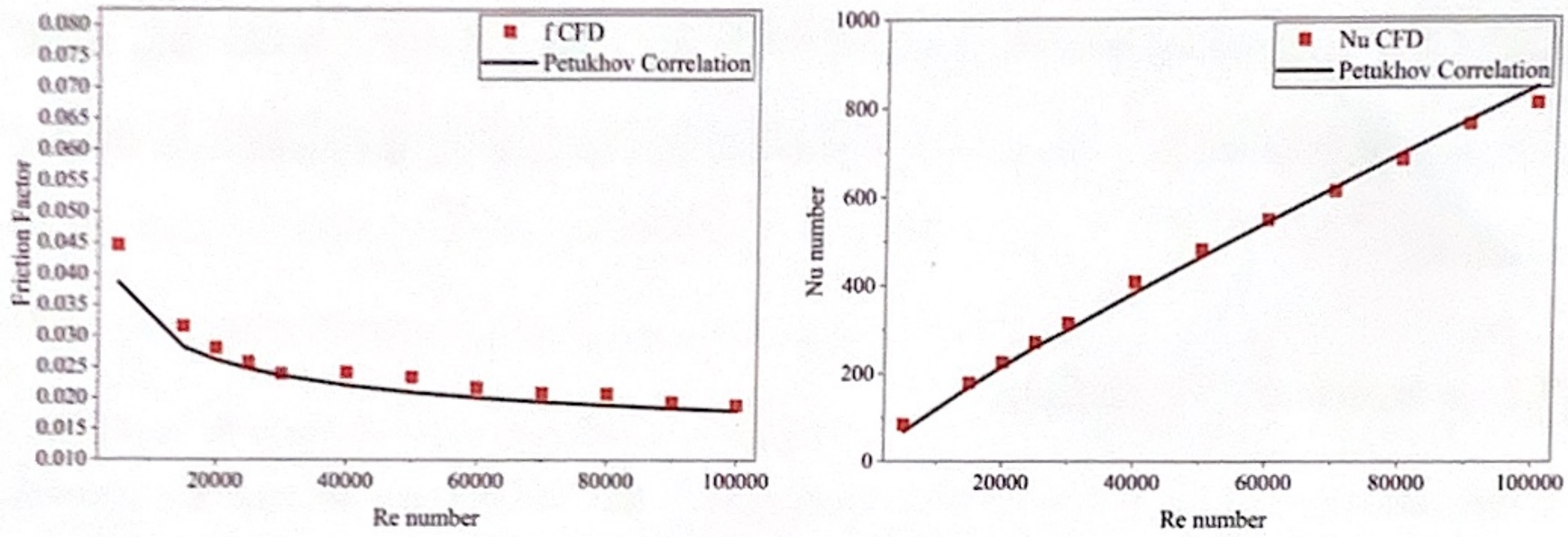


Figure 3-6: Validation of the numerical model by comparison of CFD results with empirical correlations: (a) friction factor and (b) Nusselt number.

3.7.2 Experimental Data

The thermal efficiency of a smooth receiver is compared with Dudley [76] results of his experiments to validate the present numerical model as shown in Table 3-3. This comparison showed a good agreement with the Dudley LS-2 experimental data. The absolute percentage-point differences ranged from approximately 0.30 to 1.57 percentage points, while the relative deviations were approximately 0.41–2.24%. Therefore, the model is considered suitable for baseline PTSC simulation and comparative parametric analysis.

Table 3-3: Comparison of the present numerical results with Dudley experimental data [76].

Case number	**1**	**2**	**3**	**4**
DNI (W/m^2)	933.7	982.3	937.9	920.9
Wind speed (m/s)	2.6	2.5	1	2.6
Air temperature (°C)	21.2	24.3	28.8	29.5
Flow rate (L/min)	47.7	49.1	55.5	56.8
T_{inlet} (°C)	102.2	197.5	297.8	379.5
Thermal Efficiency (Experiments)	72.51	70.17	67.98	62.34
Thermal Efficiency (Present work)	**72.21**	**68.6**	**67.19**	**61.06**

3.8 Thermophysical Properties

Before conducting the simulations, the thermophysical properties of the working fluids were examined. In the current study, Syltherm-800 is considered to be the base fluid. The hybrid nanofluids are created by suspending two different types of nanoparticles in the base fluid. As the effectiveness of the PTSC receiver is heavily dependent on the properties of the working fluid, such as its density, specific heat, thermal conductivity, and dynamic viscosity, their values were calculated using the single-phase approach because the investigated particle volume fractions are low and the objective is comparative performance screening; however, it does not capture particle migration, sedimentation, or two-phase slip.

The effective density of the hybrid nanofluid, ρ_{hnf}, was calculated from [77]:

$$\rho_{hnf} = (1-\phi)\rho_f + \phi_{p1}\rho_{p1} + \phi_{p2}\rho_{p2} \tag{3.37}$$

where ϕ is the total nanoparticle volume fraction, ϕ_{p1} and ϕ_{p2} are the volume fractions of nanoparticle 1 and nanoparticle 2, respectively, and ρ_f, ρ_{p1}, and ρ_{p2} are the densities of the base fluid and nanoparticles.

The effective specific heat of the hybrid nanofluid, $(\rho C_p)_{hnf}$, was determined as [77]:

$$(\rho C_p)_{hnf} = (1-\phi)(\rho C_p)_f + \phi_{p1}(\rho C_p)_{p1} + \phi_{p2}(\rho C_p)_{p2} \tag{3.38}$$

and therefore, the effective specific heat can be written as:

$$(C_p)_{hnf} = \frac{(1-\phi)(\rho C_p)_f + \phi_{p1}(\rho C_p)_{p1} + \phi_{p2}(\rho C_p)_{p2}}{\rho_{hnf}} \tag{3.39}$$

The effective thermal conductivity of the nanofluid was evaluated using the Maxwell model[78]:

$$k_{fnf} = k_f \left(\frac{k_{hnp} + 2k_f - 2\phi(k_f - k_{hnp})}{k_{hnp} + 2k_f + \phi(k_f - k_{hnp})} \right) \tag{3.40}$$

where k_{fnf} is the effective thermal conductivity of the hybrid nanofluid, k_f is the thermal conductivity of the base fluid, k_{hnp} is the equivalent thermal conductivity of the hybrid nanoparticles, and ϕ is the total nanoparticle volume fraction.

The effective dynamic viscosity was calculated using the Brinkman model[79]:

$$\mu_{nhf} = \mu_f (1-\phi)^{-2.5} \tag{3.41}$$

where μ_{hnf} is the dynamic viscosity of the nanofluid and μ_f is the viscosity of the base fluid.

It is important to note that the viscosity relation adopted in this study is a general model. It was selected to ensure that all investigated hybrid nanofluids were evaluated under the same modeling framework. Previous work [80] has shown that, for nanoparticle concentrations up to 3%, the use of different viscosity correlations has only a negligible effect on the predicted performance of solar collectors. Therefore, the use of this viscosity model in the present study is considered reasonable and consistent for comparative analysis.

These effective thermophysical properties were then implemented in the governing equations and used throughout the CFD simulations.

3.9 Summary

This chapter has discussed the methodology and the mathematical basis followed in the investigation regarding the thermo-hydraulic and thermodynamic efficiency of a PTSC receiver with delta-winglet vortex generators. The model depicts a generic PTSC receiver consisting of a steel-based receiver tube and a borosilicate glass enclosure with an evacuated cavity.

The fluid flow and heat transport phenomena are described by a solution of the steady, incompressible Reynolds-Averaged Navier-Stokes equations and the energy equation. The CFD solution procedure involves a pressure-based finite-volume algorithm (ANSYS FLUENT) and consists of the following steps: pre-processing (3D geometry, tetrahedral/hexahedral mesh generation), solution, and post-processing where Reynolds number, friction factor, Nusselt number, thermal efficiency, and Performance Evaluation Criterion based upon fluid and passive transport properties are evaluated alike.

Various grid independence analyses performed on a number of levels demonstrated that a computational mesh of approximately 5.7 million elements results in sensitivities in mean Nusselt number of less than 1. Validation of the CFD model was performed in two steps: first by comparison of Nusselt numbers and friction factor values in smooth tubes with Gnielinski and Petukhov equations, and in the second step by comparison of the thermal efficiency of a smooth PTSC receiver with the corresponding results of the LS-2 experimental test conducted by Dudley, with values well within CFD solution uncertainty levels and less than 1.6 % in terms of efficiency. The developed and validated CFD-optical model solution tool will form a reliable basis for the following chapters.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the numerical results for the thermal-hydraulic and thermodynamic performance of the parabolic trough solar collector receiver equipped with internal delta-winglets vortex generators and operating with either the base heat-transfer fluid or selected hybrid nanofluids. The analysis evaluates the effects of vortex-generator geometry and working-fluid thermophysical properties on flow behavior, heat-transfer enhancement, pressure-drop penalty, thermal efficiency, exergetic efficiency, and entropy generation. Since the performance of passive heat-transfer enhancement techniques depends on the balance between thermal improvement and hydraulic loss, the results are assessed using both heat-transfer indicators and overall performance criteria.

The chapter first examines the geometrical effects of the delta-winglet vortex generators through a parametric study of attack angle, blockage ratio, and pitch ratio. The results are discussed in terms of heat-transfer enhancement, friction behavior, and performance evaluation criterion (PEC) in order to identify the optimum configuration. To support the interpretation of the numerical results, temperature, velocity, and turbulent kinetic energy contours are also analyzed to explain the flow redistribution, boundary-layer disturbance, and mixing mechanisms produced by the inserts.

After identifying the optimum vortex-generator geometry, the chapter investigates the use of hybrid nanofluids as enhanced working fluids. Different hybrid nanofluid types are first screened under the optimized receiver configuration, followed by further optimization of the selected nanofluid through variation of nanoparticle volume concentration and mixing ratio. In this way, the chapter provides a complete assessment of how geometric modification and fluid-property enhancement, individually and in combination, improve the performance of the PTSC receiver.

4.2 Geometrical impact

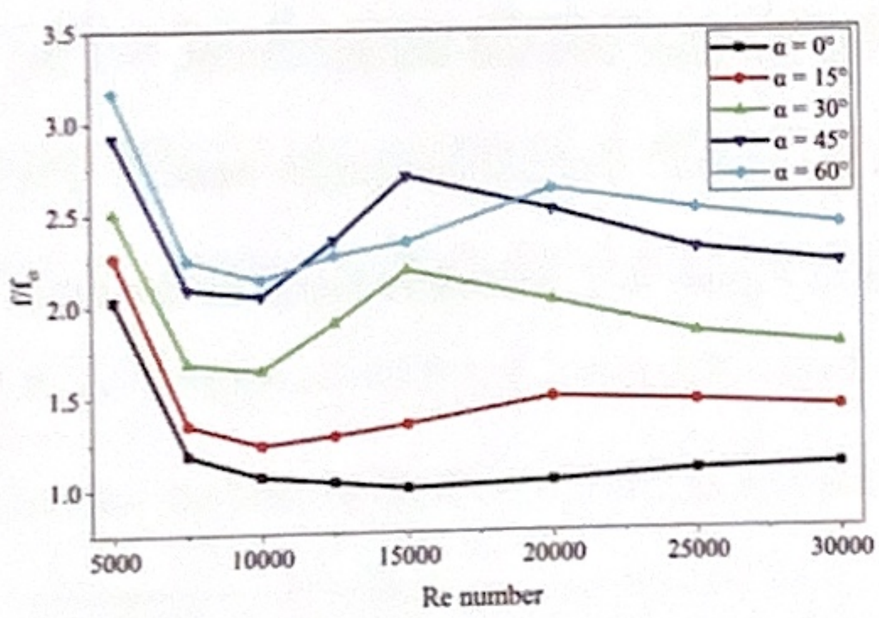
The optimization was performed in three stages: first, the attack angle was varied for several blockage ratios at a fixed pitch ratio; second, the blockage ratio was screened at the selected attack angle; and third, the pitch ratio was varied at the selected attack angle and blockage ratio. In all cases, the design decision was based on PEC rather than on heat-transfer enhancement alone.

4.2.1 Effect of Attack Angle

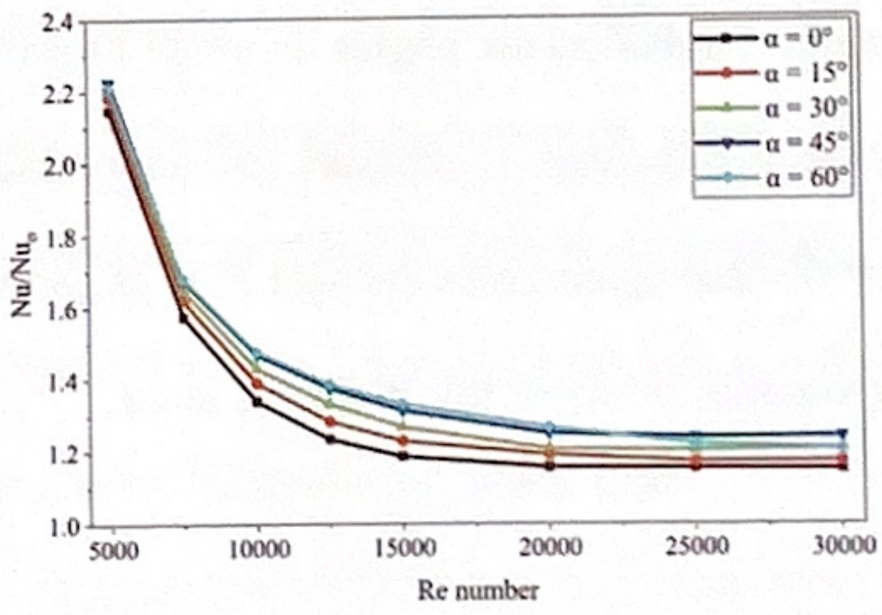
In this section, the influence of the vortex-generator attack angle, α, on the thermal-hydraulic performance of the receiver tube is examined. The attack angle was varied in the range from 0° to 60° while the pitch ratio was kept constant at $PR = 1.6$, and the analysis was repeated for three blockage ratios, $BR = 0.1$, 0.2 and 0.3, in order to isolate the effect of winglet orientation under different blockage conditions. The purpose of this study is to determine how changing the inclination of the vortex generators modifies the strength of the induced secondary flow, the disturbance of the boundary layer, and the associated pressure-drop penalty. It is expected that increasing α will intensify flow mixing and

improve heat transfer due to stronger vortex formation. However, it is also expected to increase hydraulic losses because the projected area facing the flow becomes larger. Therefore, this section aims to identify the angle that provides the most favorable balance between thermal enhancement and friction penalty.

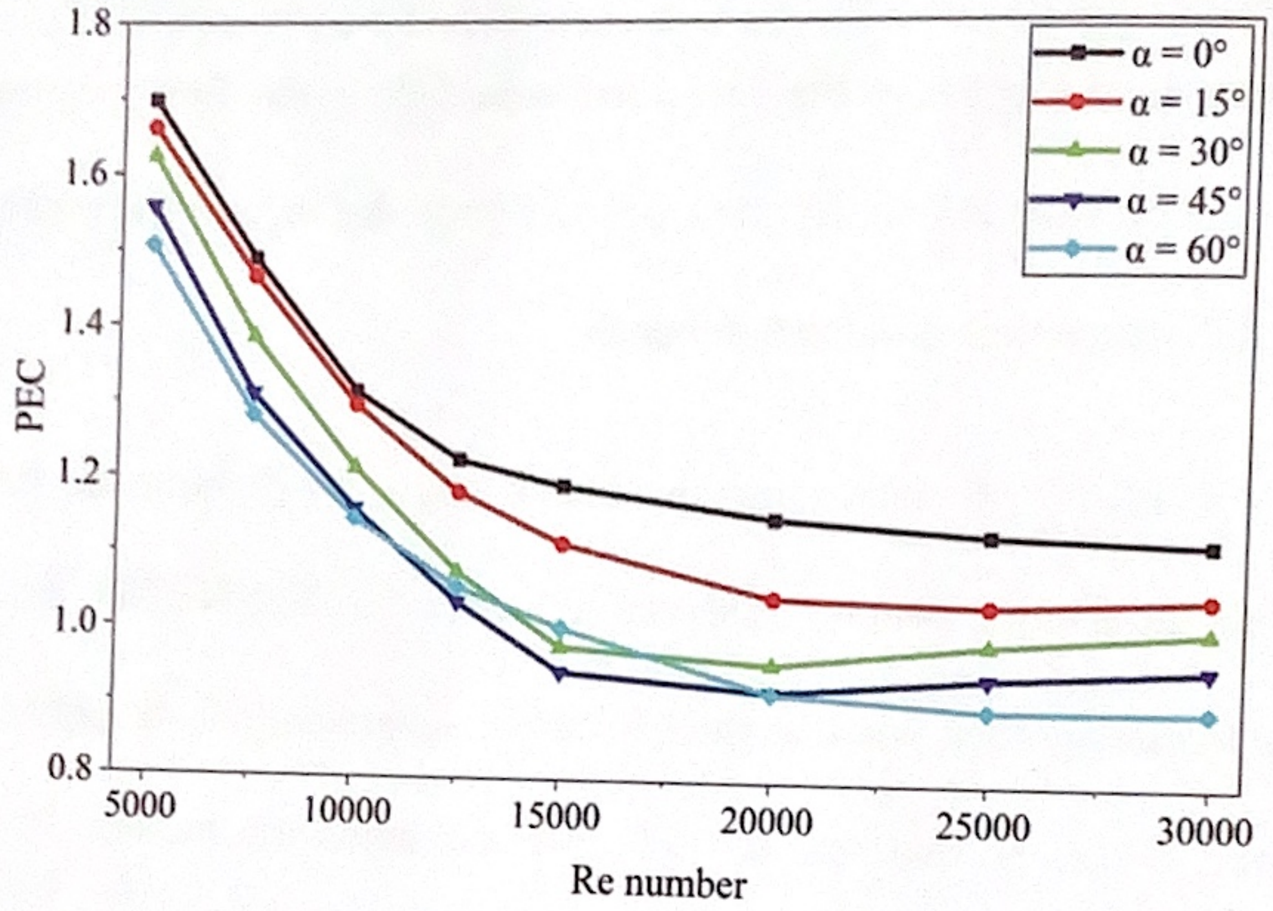
At BR = 0.1 and PR = 1.6, increasing the attack angle raises both the heat-transfer enhancement and the friction penalty as shown in Figure 4-1. The Nu/Nu0 curves remain relatively close to one another, whereas the f/f0 curves separate more strongly as the angle increases. Figure 4-1(c) present that $\alpha = 0°$ case maintains the highest PEC over the investigated Reynolds-number range and is therefore the preferred angle at this blockage ratio.



(a)



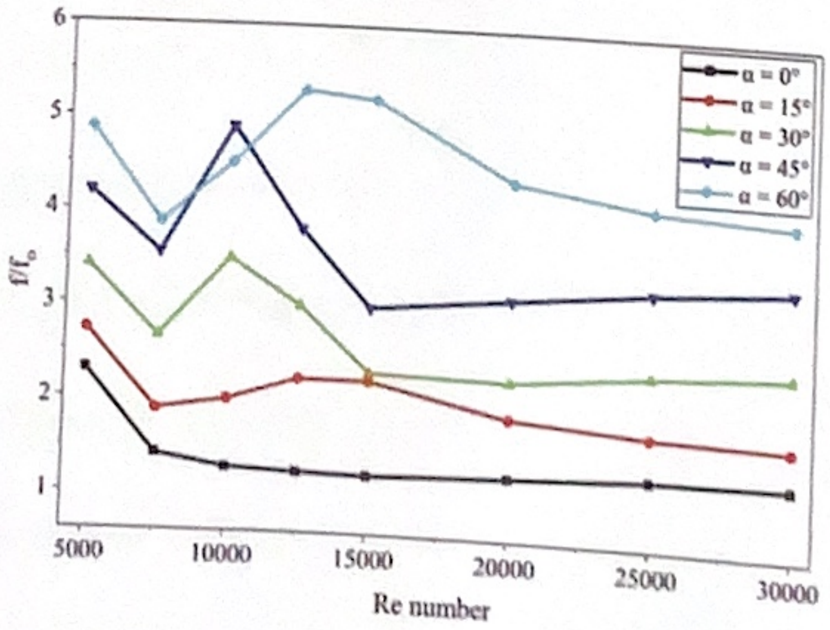
(b)

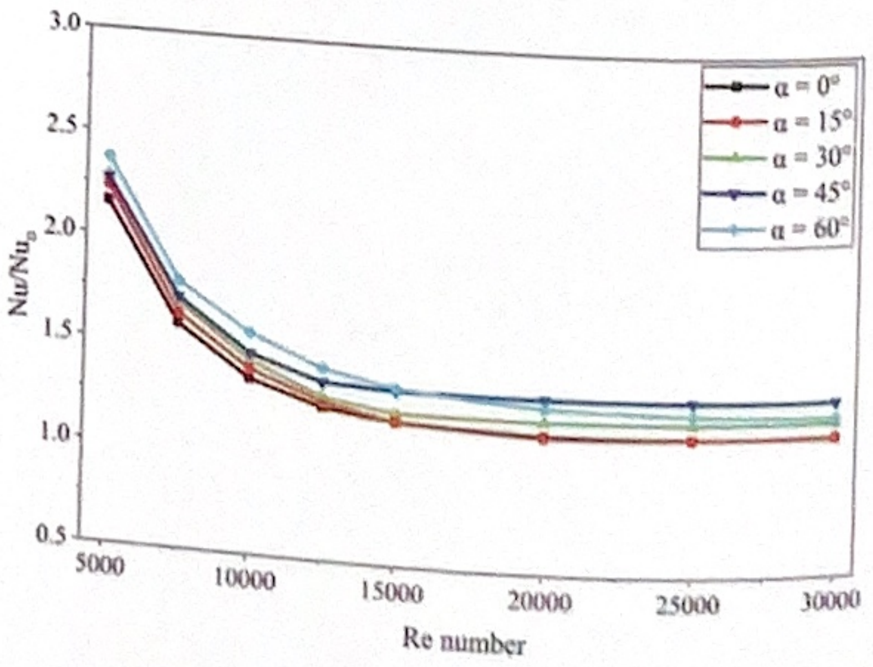


(c)

Figure 4-1: Effects of attack angle on thermal-hydraulic performance for $BR = 0.1$ and $PR = 1.6$: (a) friction-factor ratio, (b) Nusselt-number ratio, and (c) performance evaluation criterion (PEC).

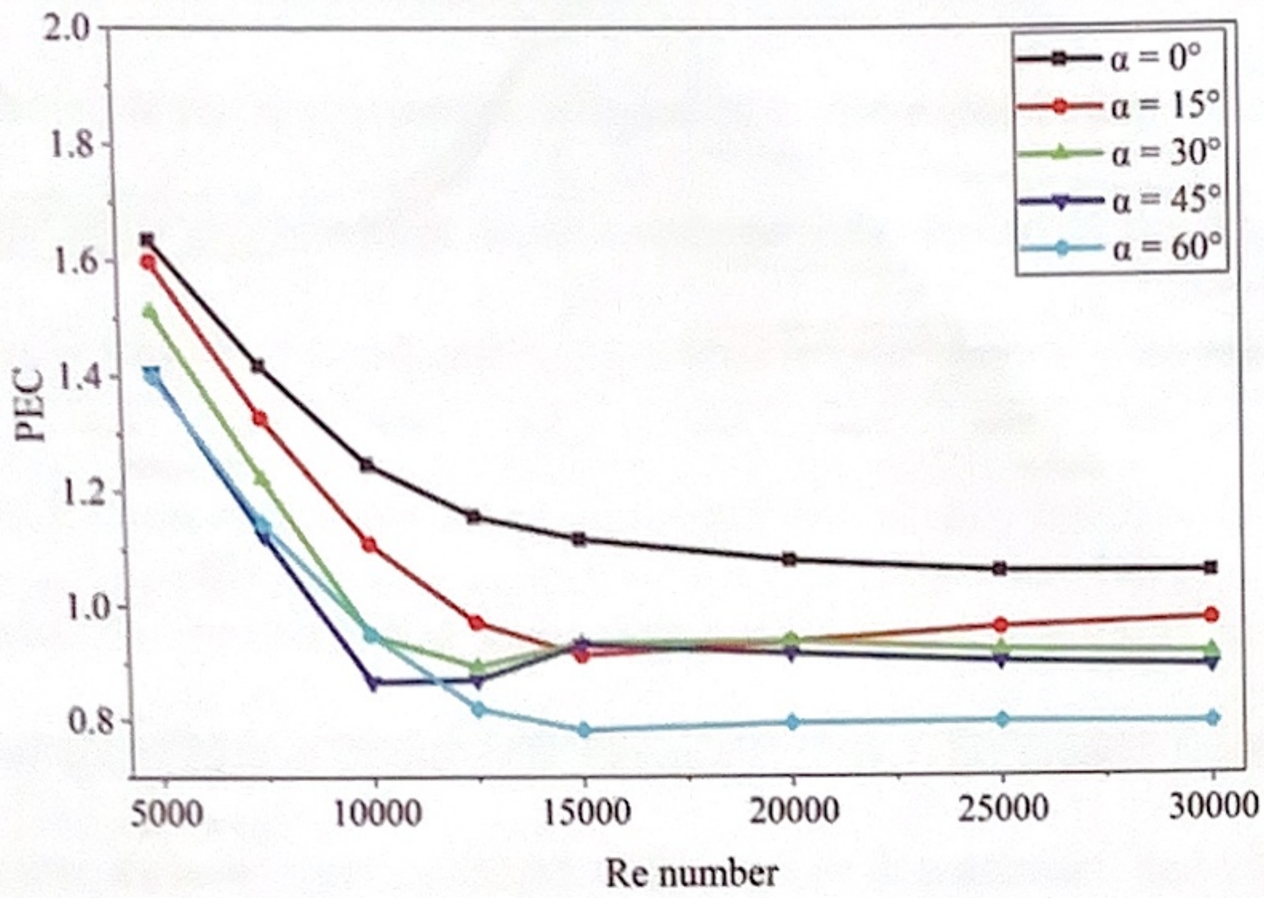
When BR is increased to 0.2, the hydraulic penalty becomes more pronounced for all angles. Higher attack angles continue to provide stronger thermal enhancement, but the associated increase in pressure loss suppresses the overall thermal-hydraulic benefit. The $\alpha = 0°$ case again gives the best PEC as shown in Figure 4-2, therefore remains the most favourable choice at this blockage level.





(a)

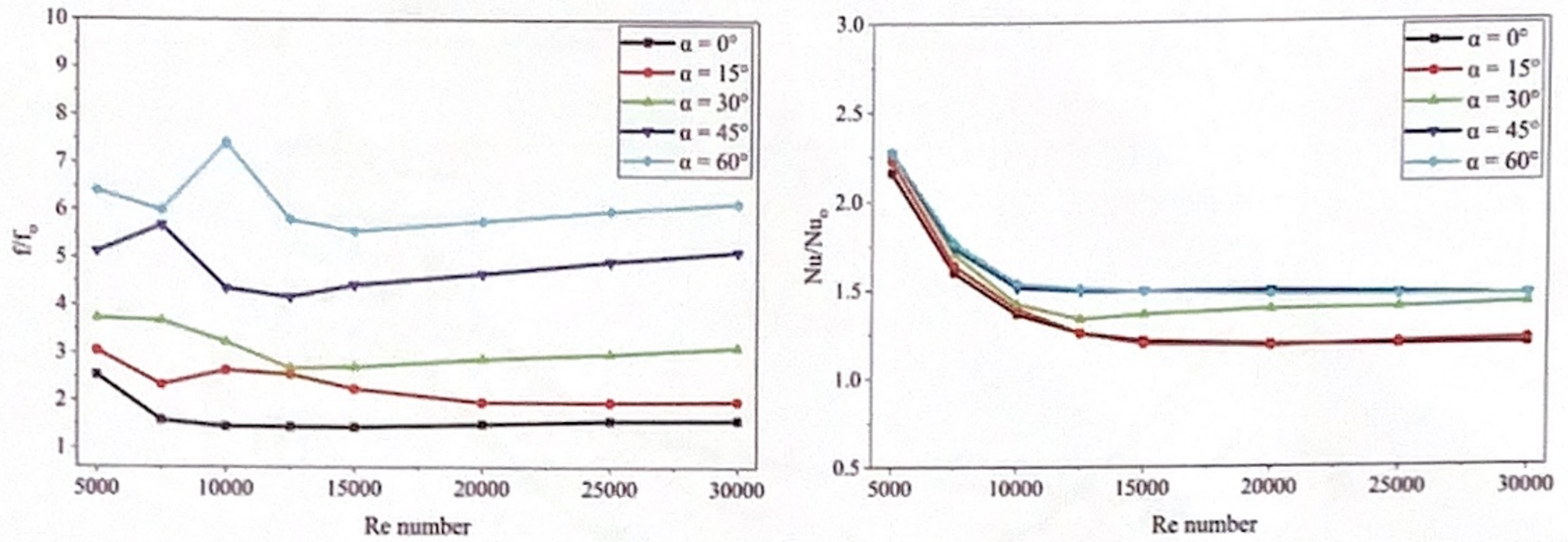
(b)



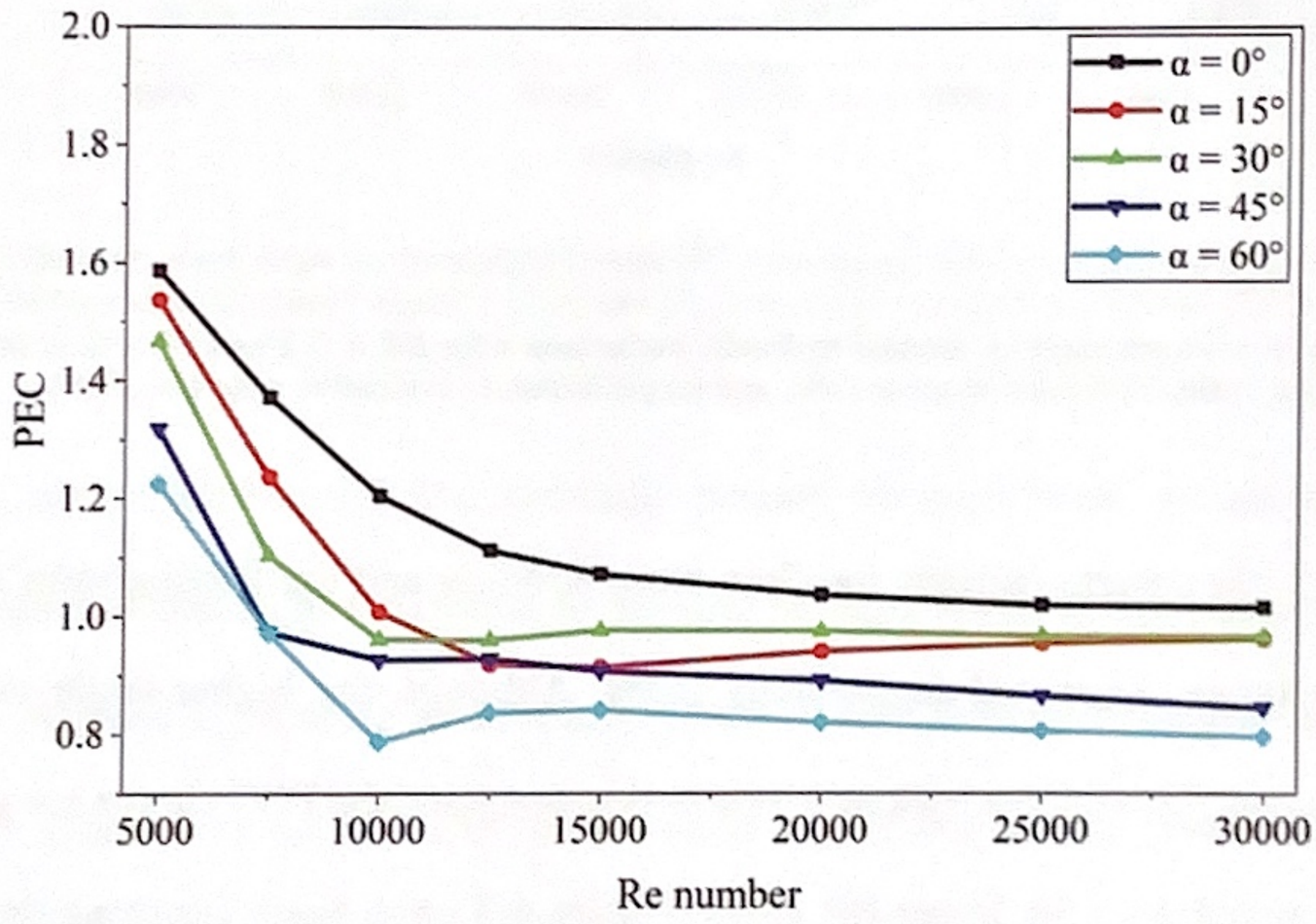
(c)

Figure 4-2: Effects of attack angle on thermal-hydraulic performance for $BR = 0.2$ and $PR = 1.6$: (a) friction-factor ratio, (b) Nusselt-number ratio, and (c) performance evaluation criterion (PEC).

For BR = 0.3, the insert obstructs the flow more strongly and the friction ratio becomes significantly larger compared to previous cases. Although the higher-angle cases still produce elevated Nu/Nu0, the hydraulic cost is dominant and the PEC curves are generally reduced compared with the lower-BR cases. Figure 4-3 once again confirms that $\alpha = 0°$ provides the best overall compromise and is retained as the preferred attack angle.



(a) (b)



(c)

Figure 4-3: Effects of attack angle on thermal-hydraulic performance for $BR = 0.3$ and $PR = 1.6$: (a) friction-factor ratio, (b) Nusselt-number ratio, and (c) performance evaluation criterion (PEC).

The attack-angle study demonstrated that the main observable in the graphs is that changing α modifies both the heat-transfer enhancement and the friction penalty in a systematic way, as angle increases the winglet surface becomes more exposed to the incoming flow. This produces stronger transverse motion and more intense streamwise vortices, so the fluid

near the heated wall is swept more effectively toward the core, while colder core fluid is pushed back toward the wall. Because of this mechanism, the heat-transfer curves generally improve as the angle increases from the aligned case toward the inclined cases. In physical terms, a larger angle and blockage cause a stronger interruption of the thermal boundary layer, and this is why the Nusselt-ratio curves tend to rise when α increases.

However, the friction curves rise at the same time, and usually much faster than the heat-transfer curves. The reason is that increasing α does not only strengthen the vortex, but also increases the effective frontal area facing the flow. This creates stronger stagnation zones upstream of the winglets, stronger wake regions downstream, and larger form drag. The flow then loses more mechanical energy as it passes each insert. Thus, although the thermal field becomes more mixed, the pumping-power requirement rises sharply.

The friction behavior at low Re number deserves special attention. At low Re, the core momentum of the flow is lower, so the fluid has less ability to pass through the obstruction smoothly. Under these conditions, the VG-induced wake and recirculation regions occupy a relatively larger portion of the tube cross section, and the added drag becomes more pronounced.

The reason for the abnormally high friction factor ratio at α=30° and 45°, especially around Re=15000, is the coincidence of the buoyancy-driven secondary vortices and the flow orientation formed due to the VGs. At those intermediate angles of attack, the force due to gravity is large enough to influence the longitudinal vortices formed due to the vortex generators, increasing the rotational effect and enhancing the mixing near the wall. At Re=15000, the flow experiences a state where the vortices are highly developed and

energetic but still have enough coherence until being affected by inertial dominance at high Reynolds numbers. The favorable coincidence leads to an increased interaction between the secondary flow formations and the wall boundary layer, causing higher wall shear stress and hence a significant increase in the friction factor ratio. The dominance of inertial forces at high Reynolds numbers makes the effect of buoyancy-driven vortex interaction insignificant, explaining the gradual reduction in the f/f0 ratio after that point.

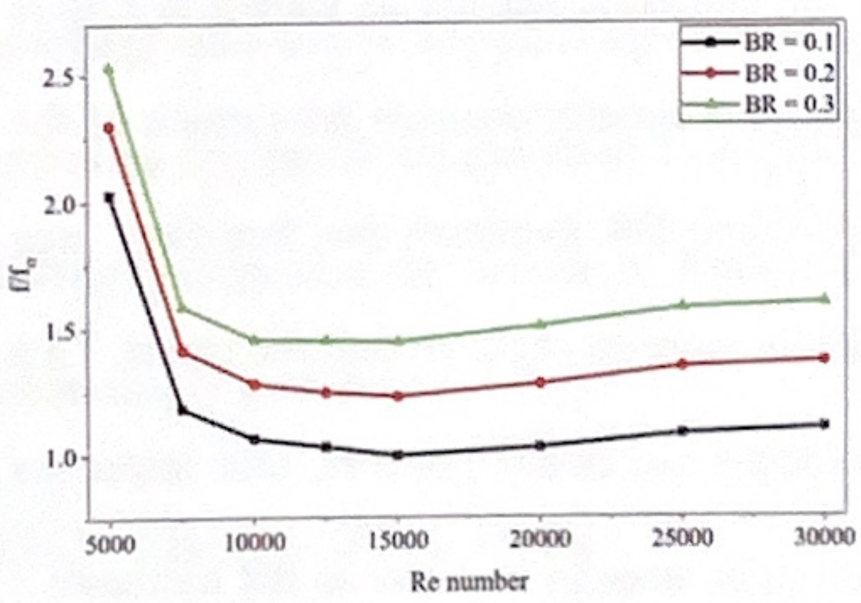
The selection of $\alpha = 0°$can therefore be justified physically. Even at zero angle, the insert still disturbs the near-wall flow because the support structure and winglet geometry remain present inside the tube. This disturbance is enough to generate beneficial mixing and boundary-layer thinning, but without the very large form drag associated with inclined winglets. Once the angle is increased, the extra heat-transfer gain becomes smaller than the extra pressure-drop penalty. Therefore, the graphs indicate that $\alpha = 0°$gives the best overall compromise, not because it produces the strongest mixing, but because it produces the most economical mixing.

These trends are consistent with previous PTSC investigations on internal fins and vortex-generator inserts [28],[31], which reported that configurations producing stronger flow disturbance do not necessarily provide the best overall performance, because the corresponding increase in drag and pressure drop can outweigh the thermal gain.

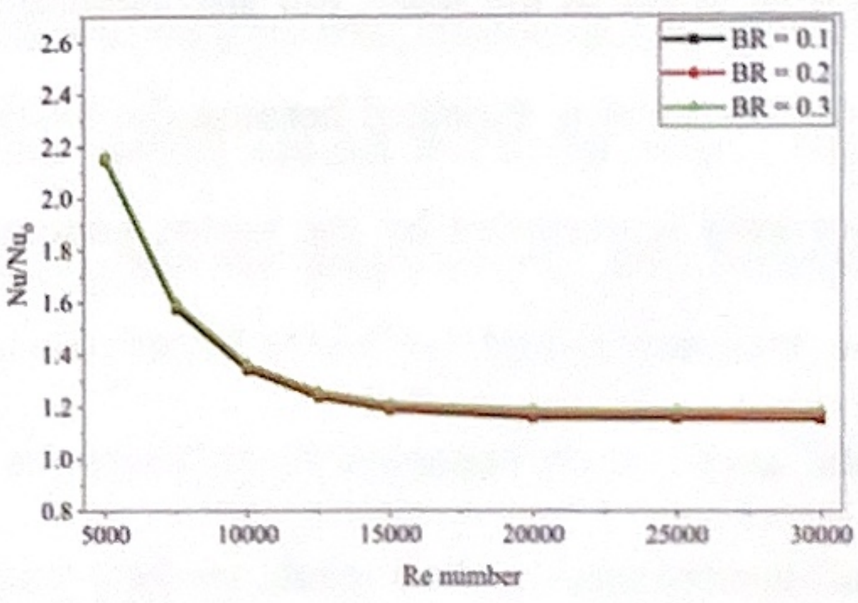
4.2.2 Effect of Blockage Ratio

This section investigates the effect of blockage ratio, BR, on the performance of the receiver tube with vortex generators. The variation of the blockage ratio is studied under the condition of keeping the attack angle constant at $\alpha = 0°$ and the pitch ratio was fixed

at $PR = 1.6$. This investigation aims to assess the influence of the size of the vortex generators on its internal flow structure, boundary-layer separation, and thermal-hydraulic performance. A higher blockage ratio is expected to produce stronger vortices and more radial mixing, which may enhance heat transfer. At the same time, increasing the blockage ratio reduces the available flow area and increases flow obstruction, which is expected to raise the pressure drop significantly. Accordingly, this section is intended to determine the blockage level that offers sufficient thermal improvement without causing an excessive hydraulic penalty. The results shown in Figure 4-4 that a larger BR increases both heat-transfer enhancement and pressure drop; however, the friction penalty rises significantly compared to the thermal gain. Accordingly, BR = 0.1 delivers the highest PEC and is selected as the optimum blockage ratio within the investigated range.



(a)



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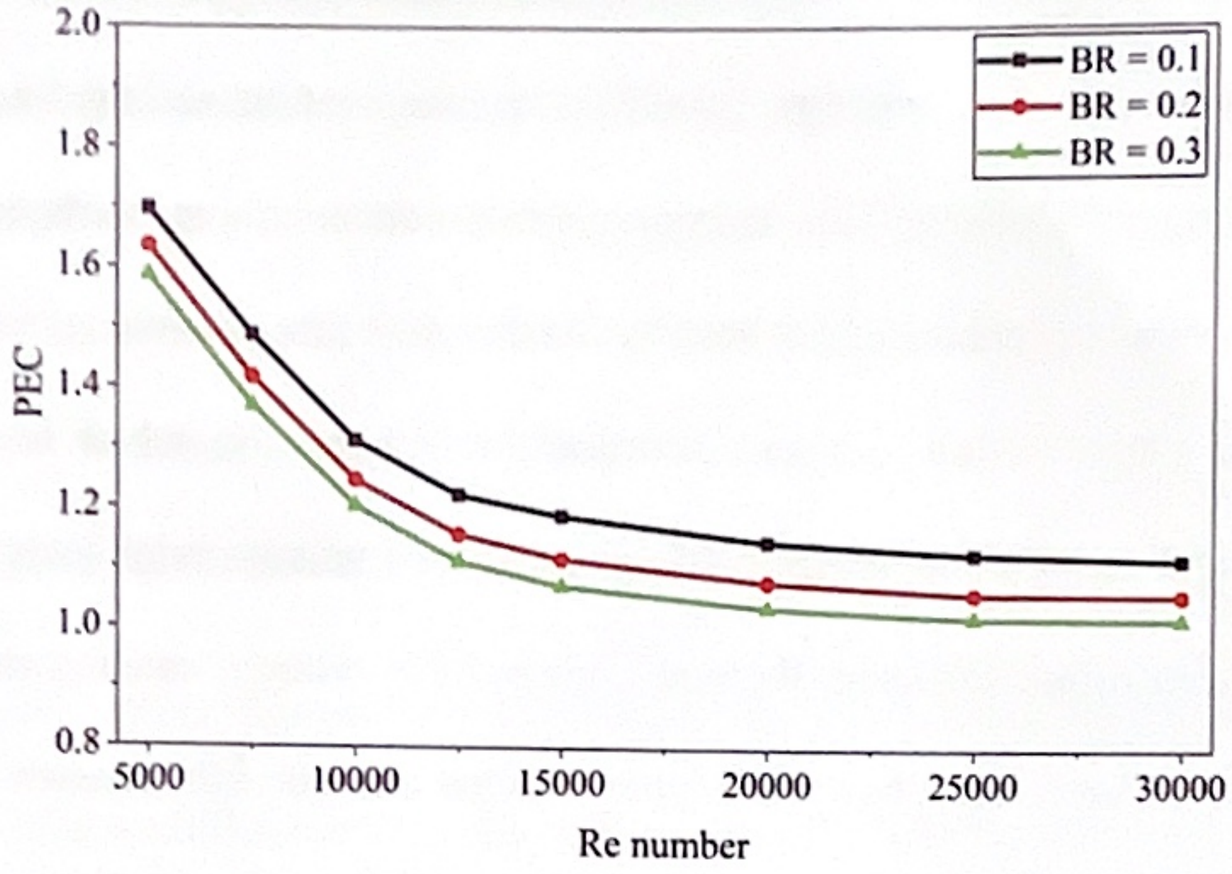


(c)

Figure 4-4: Effects of blockage ratio on thermal-hydraulic performance for $\alpha = 0^{\circ}$ and $PR = 1.6$: (a) friction-factor ratio, (b) Nusselt-number ratio, and (c) performance evaluation criterion (PEC).

For the blockage-ratio study, the main observable is that increasing BR strengthens the thermal effect of the insert but also intensifies the hydraulic penalty as shown in Figure 4-4. This trend is expected because the blockage ratio directly controls how much of the flow area is occupied by the vortex generator. When BR increases the free-flow area becomes smaller and the fluid is forced to accelerate more strongly around the insert. This local acceleration enhances shear, promotes stronger secondary vortices, and improves wall-to-core mixing. As a result, the heat-transfer ratio tends to improve as BR increases.

At the same time, larger blockage causes a much stronger penalty in friction. Physically, this happens because a larger VG obstructs more of the cross section, producing a larger adverse pressure region, stronger separation behind the insert, and a thicker wake. In addition, stronger vortices dissipate more energy into turbulence. Therefore, the hydraulic cost rises faster than the thermal benefit. This is exactly why larger vortex generators are not automatically better, even though the contours show more intense flow activity as

shown in Figure 4-6. Stronger disturbance is beneficial only up to the BR = 0.1 where the heat transfer gained still compensates for the extra pumping power. Beyond that point, the insert becomes over-aggressive.

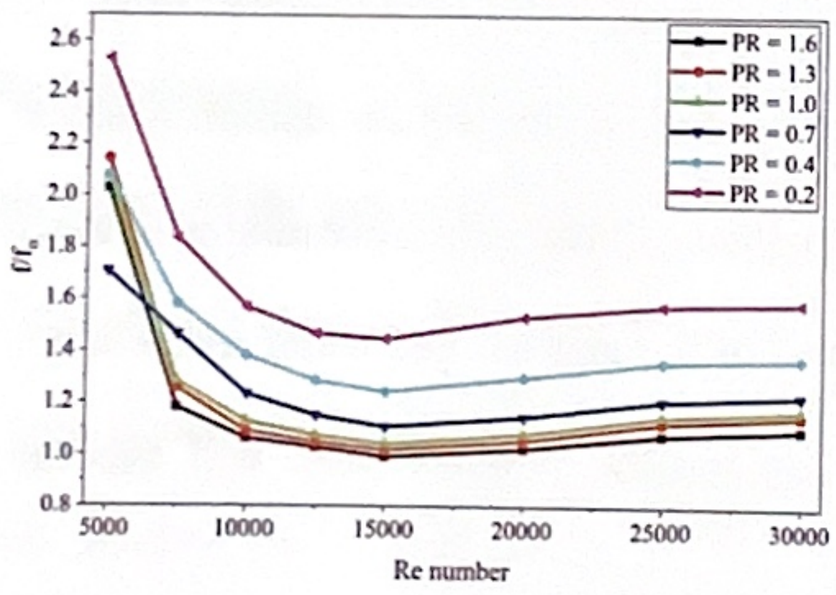
The low-Re friction behavior is important here. At lower Reynolds numbers, the relative obstruction caused by a larger blockage ratio is especially severe because the flow inertia is weaker. The fluid cannot recover quickly after passing the insert, so the wake and low-momentum regions remain strong over a larger distance. This makes the effect of the friction factor very clear in the case of low Re numbers. In high Re numbers, there will be more kinetic energy available to restore the flow; hence, the relative drop will seem smoother even though the pressure drop remains high.

The value of BR = 0.1 is highly rational and realistic from a physical standpoint. A lower blockage ratio will be adequate for disturbing the boundary layer to achieve mixing without imposing too much resistance to flow. BR values of 0.2 and 0.3 had better temperature distribution profile as shown in Figure 4-6 (c), but the pressure drop will become increasingly prohibitive.

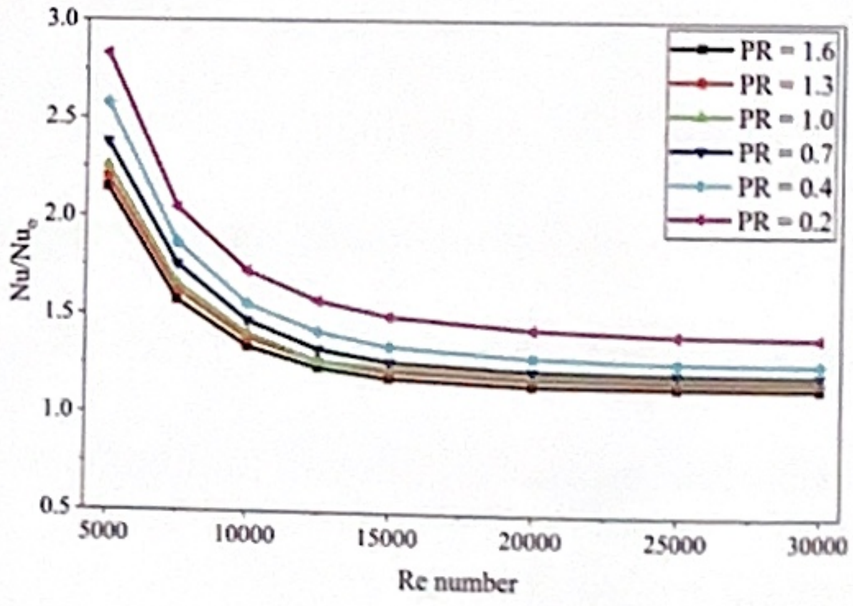
4.2.3 Effect of Pitch Ratio

In this section, the influence of pitch ratio, PR, on the receiver performance is analyzed. The pitch ratio was varied while keeping the blockage ratio constant at $BR = 0.1$ and the attack angle fixed at $\alpha = 0°$, so that only the effect of the spacing between successive vortex generators could be evaluated. The main purpose of this study is to understand how the longitudinal distance between inserts controls the frequency of flow disturbance and the redevelopment of the thermal and hydrodynamic boundary layers along the tube. A

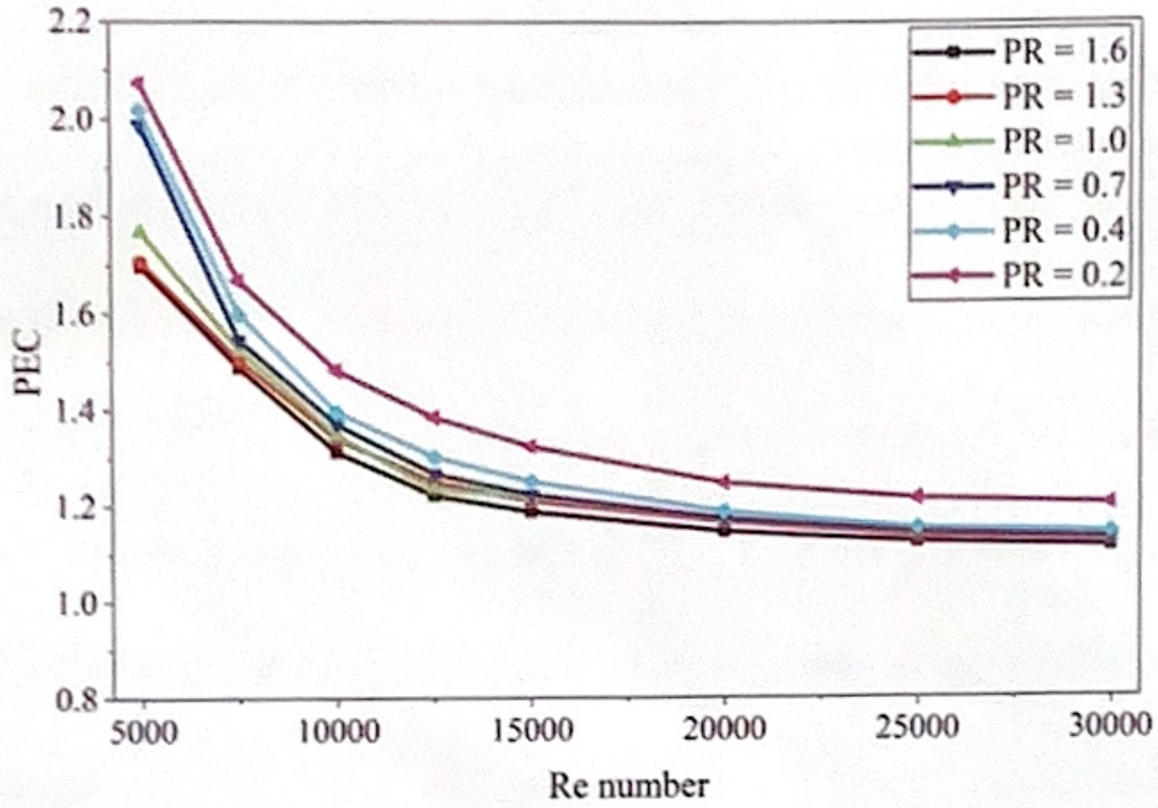
smaller pitch ratio is expected to maintain more frequent interruption of the boundary layer and therefore improve heat transfer, since the fluid is repeatedly mixed before the boundary layer fully redevelops. However, closer spacing also means a larger number of inserts per unit length, which is expected to increase the cumulative pressure-drop penalty. Thus, this section seeks to identify the pitch ratio that provides the most effective compromise between repeated thermal enhancement and the associated increase in hydraulic resistance. Figure 4-5 illustrates the results of pitch ratio study, among the tested values, PR = 0.2 gives the highest PEC and selected as best-performing case within the investigated range.



(a)



(b)



(c)

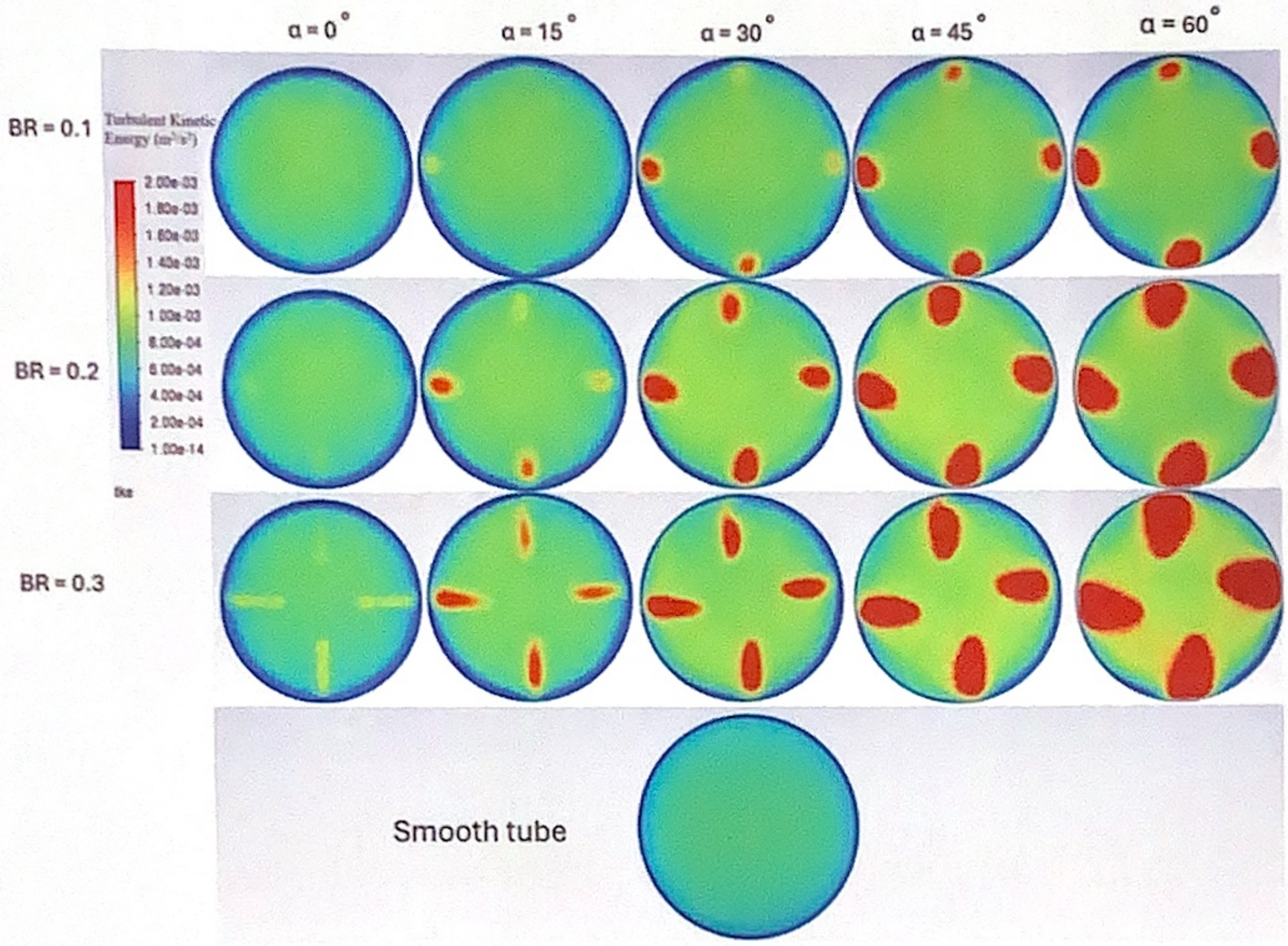
Figure 4-5: Effects of pitch ratio on thermal-hydraulic performance for $\alpha = 0^{\circ}$ and $BR = 0.1$: (a) friction-factor ratio, (b) Nusselt-number ratio, and (c) performance evaluation criterion (PEC).

The key observable in Figure 4-5 is that reducing PR, meaning reducing the spacing between successive vortex generators, changes both the frequency of flow disturbance and the redevelopment length of the boundary layer. A large pitch ratio means that the inserts are far apart, so after each vortex generator the boundary layer has more distance to recover and grow again before the next insert is reached. Under such conditions, part of the enhancement effect is lost between two consecutive elements. This is why the heat-transfer improvement is generally weaker at large pitch ratio.

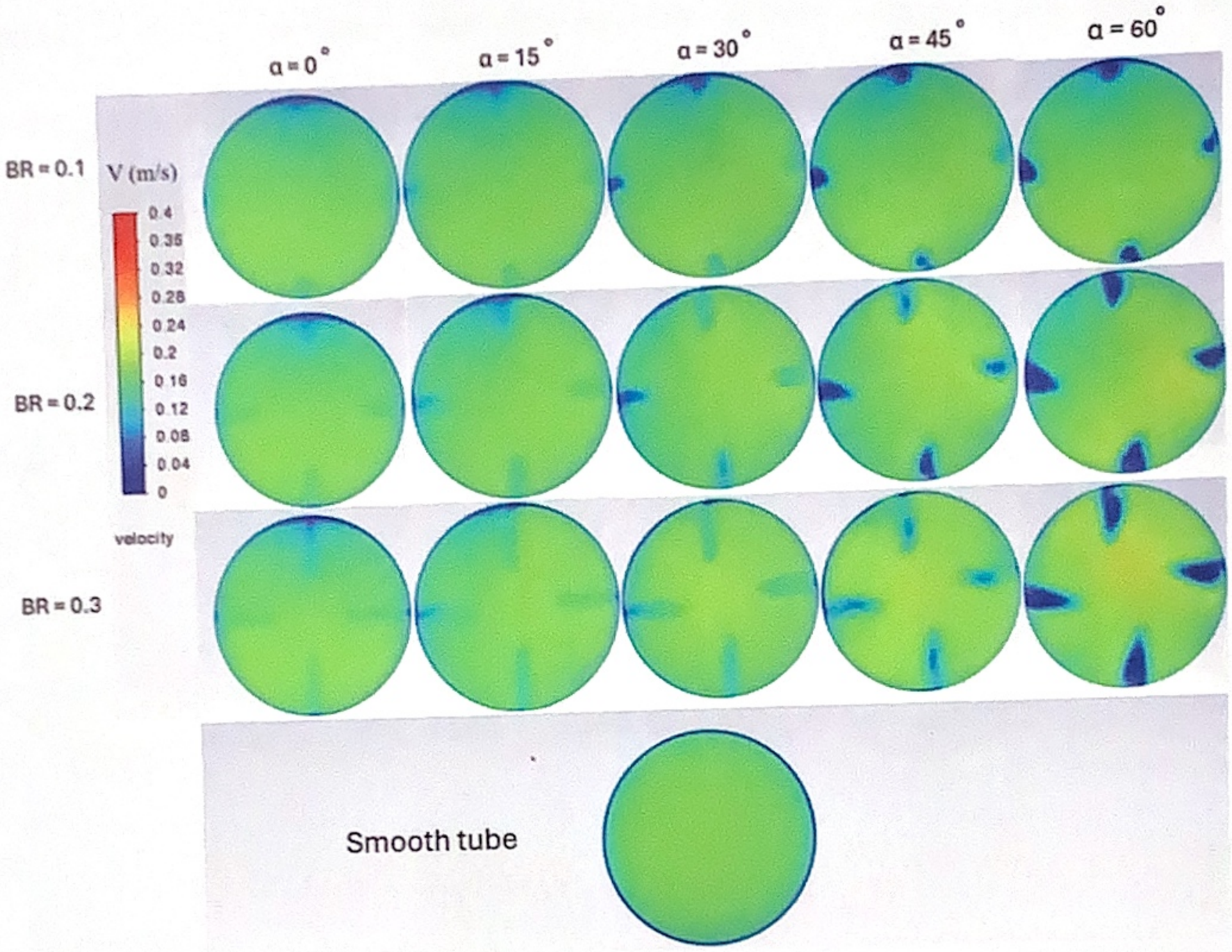
When PR decreases, the inserts are placed closer together, so the boundary layer is interrupted more frequently. The fluid near the wall is renewed repeatedly, and the secondary-flow effect is maintained over a larger portion of the tube length. This explains why the heat-transfer graphs improve as the pitch ratio becomes smaller. In physical terms, a short pitch prevents full thermal redevelopment and keeps the flow in a continuously disturbed state, which is favorable for convective heat transfer.

Naturally, reducing the pitch ratio also increases friction because more inserts are encountered over the same tube length. Each insert contributes an additional local loss, so the cumulative pressure drop becomes larger as PR decreases. This behavior is aligned with Liu et al. study [81], where pitch ratio was identified as a key geometric parameter governing the balance between heat-transfer enhancement and frictional loss. However, in the best performing configuration (PR = 0.2) the benefit from repeated boundary-layer disruption is greater than the added hydraulic penalty within the tested range. This is an important result because it shows that the chosen low-angle, low-blockage geometry is hydraulically mild enough that it can be repeated more frequently without making the pressure loss unacceptable. If the insert had been more aggressive, such as at larger α or larger BR, then a very small pitch would probably have become unfavorable.

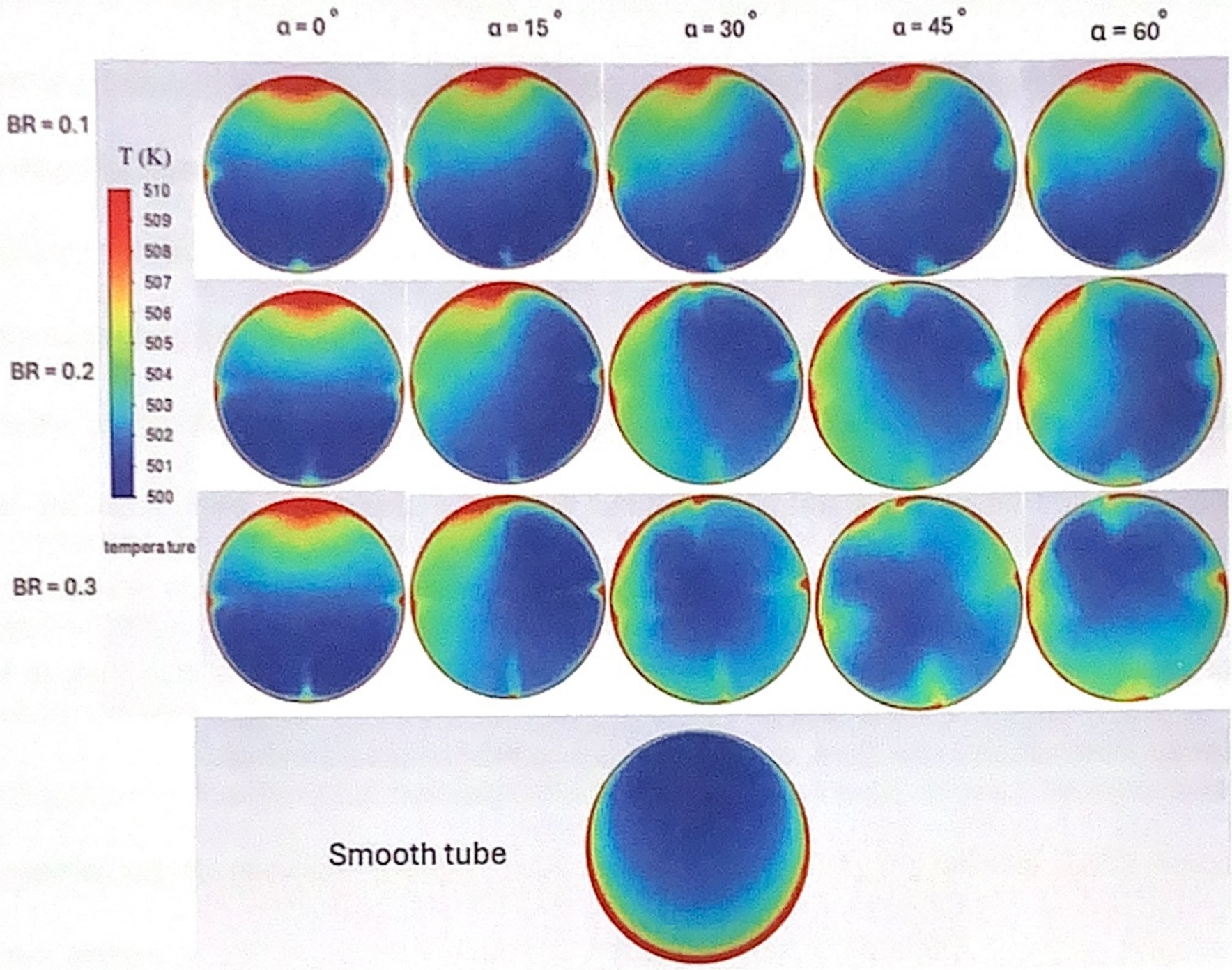
The variations of temperature, velocity, and TKE contours as shown in Figure 4-6, reveal the reasons behind the impact of vortex-generator geometries on flow patterns and thermal behavior inside the receiver tube. The cases examined consist of attack angles of $\alpha = 0°, 15°, 30°$, $45°$, and $60°$ along with blockage ratios $BR = 0.1, 0.2,$and 0.3. Overall, the increase of either blockage ratio or the attack angle increases the disturbance level imposed on the core flow. This can be observed from the contours because the increase in disturbance results in more developed zones of secondary flows, higher TKE levels, and more distorted temperature distribution as shown in Figure 4-6. All these prove that the vortex-generators increase mixing between the core flow and near-wall flows that is useful for improving heat exchange. At the same time, the contours show that strong mixing does not necessarily guarantee good performance.


α = 0°
α = 15°
α = 30°
α = 45°
α = 60°
BR = 0.1
BR = 0.2
BR = 0.3
Turbulent Kinetic Energy (m²/s²)
2.00e-03
1.80e-03
1.60e-03
1.40e-03
1.20e-03
1.00e-03
8.00e-04
6.00e-04
4.00e-04
2.00e-04
1.00e-14
Smooth tube

(a)


α = 0°
α = 15°
α = 30°
α = 45°
α = 60°
BR = 0.1
V (m/s)
0.4
0.36
0.32
0.28
0.24
0.2
0.16
0.12
0.08
0.04
0
velocity
BR = 0.2
BR = 0.3
Smooth tube

(b)



(c)

Figure 4-6: (a) Cross-sectional temperature contours, (b) velocity contours, and (c) turbulent kinetic energy contours for different vortex-generator configurations.

From the TKE contours shown in Figure 4-6 (a), it is observed that the magnitude and spatial extent of turbulence increase progressively as the attack angle increases from 0°to 60°, and as the blockage ratio increases from 0.1 to 0.3. At low blockage ratio and zero attack angle, the TKE distribution is relatively mild and localized, indicating a controlled disturbance of the flow and this configuration gave best performance. When either α or BR increases, high-TKE regions become more intense and occupy larger portions of the cross section. This means that the increased intensity of the vortex generated by the bigger VG leads to an increase in turbulence production and dissipation, resulting in higher pressure losses.

The velocity contours shown in Figure 4-6 (b), show that for $BR = 0.1$and $\alpha = 0°$, the cross-sectional velocity field remains comparatively uniform, with only limited low-velocity disturbance near the VG influence zones. This indicates that the insert promotes mixing without severely obstructing the bulk flow. As the blockage ratio increases to 0.2 and 0.3, and as the angle increases to 45°and 60°, the low-velocity regions become more pronounced and elongated. These zones indicate stronger flow deceleration, local separation, and larger wake formation behind the vortex generators. Such behavior is undesirable from a hydraulic viewpoint because it increases drag and flow resistance. Therefore, while larger VGs create a more energetic flow structure, they also impose a stronger obstruction to the fluid, causing a significant rise in pressure drop.

Recent PTSC studies [31], [82], also indicate that stronger vortices are not always thermally optimal. Although insert inclination and more intense vortex structures can increase mixing, vortex interaction and excessive flow distortion may also increase drag and consume turbulent energy inefficiently.

The temperature contours shown in Figure 4-6 (c), reveal the thermal consequence of this flow behavior. At higher blockage ratios and attack angles, the temperature field becomes more distorted and more mixed across the tube cross section, indicating stronger transport of thermal energy from the heated wall toward the fluid core. This confirms that larger VGs can improve thermal mixing and reduce thermal stratification. However, the thermal improvement becomes progressively less economical as the insert size and angle increase. The additional heat gained by stronger mixing is not proportional to the hydraulic penalty introduced.

This explains why larger vortex generators are not the optimum configuration, even though their contours appear to show better mixing. Increasing the blockage ratio and attack angle increases the intensity of vortices, generation of turbulence, and heat transport. However, it also increases the strength of the recirculation regions, blocking of flow, and energy losses due to both viscosity and shape. Consequently, the pressure loss increase is the dominating factor that offsets the advantage gained by increased heat transfer.

Accordingly, the configuration with $BR = 0.1$and $\alpha = 0°$is identified as the best design within explored cases. This case provides sufficient flow disturbance to enhance the heat transfer process while maintaining the lowest pressure-drop penalty among the tested configurations results in the best performance owing to its good balance between heat transfer and hydraulic requirements.

4.2.4 Optimum Configuration and Work Summary

The screening process as summarized in Table 4-1 leads to a clear design recommendation. The attack-angle study consistently favored $\alpha = 0°$ across all investigated blockage ratios. The blockage-ratio comparison then identified BR = 0.1 as the best thermal-hydraulic compromise, and the pitch-ratio study showed that PR = 0.2 produced the highest overall PEC under the selected α and BR values. Figure 4-7 shows 3D view of the best design selected.

Table 4-1: Summary of the parameter screening process and selected values.

Design variable	Values investigated	Selected value	Basis for selection
Attack angle α	0°, 15°, 30°, 45°, 60°	0°	Highest PEC across the evaluated BR cases
Blockage ratio BR	0.1, 0.2, 0.3	0.1	Best balance between heat-transfer gain and pressure-drop penalty
Pitch ratio PR	1.6, 1.3, 1.0, 0.7, 0.4, 0.2	0.2	Highest PEC in the final optimization stage

- Increasing attack angle and blockage ratio can improve heat-transfer enhancement, but the accompanying pressure-drop penalty reduces the overall thermal-hydraulic benefit.
- Reducing pitch ratio improves the combined performance of the insert over the tested range, and PR = 0.2 provides the strongest PEC response.
- The recommended design for subsequent detailed analysis is α = 0°, BR = 0.1, and PR = 0.2.

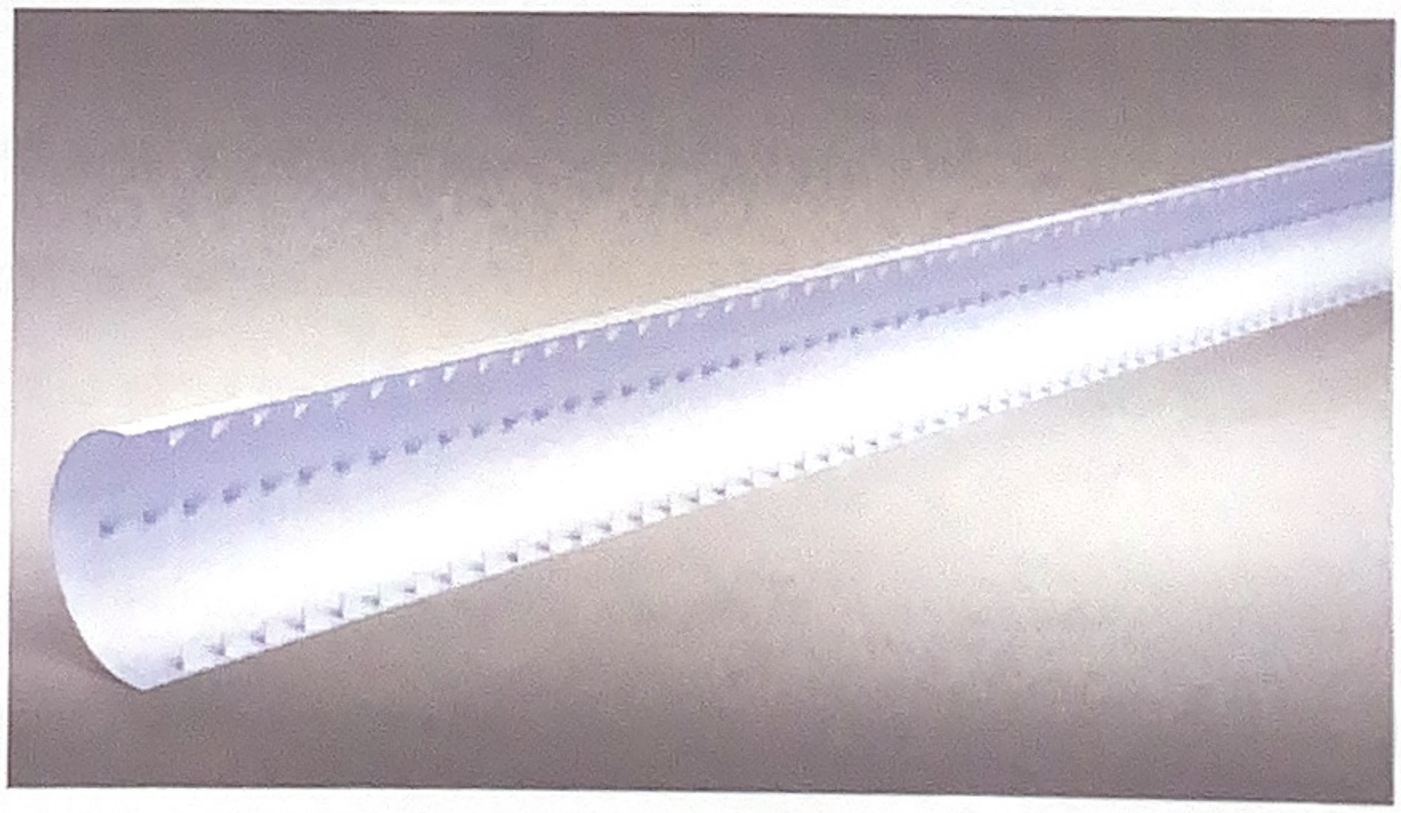
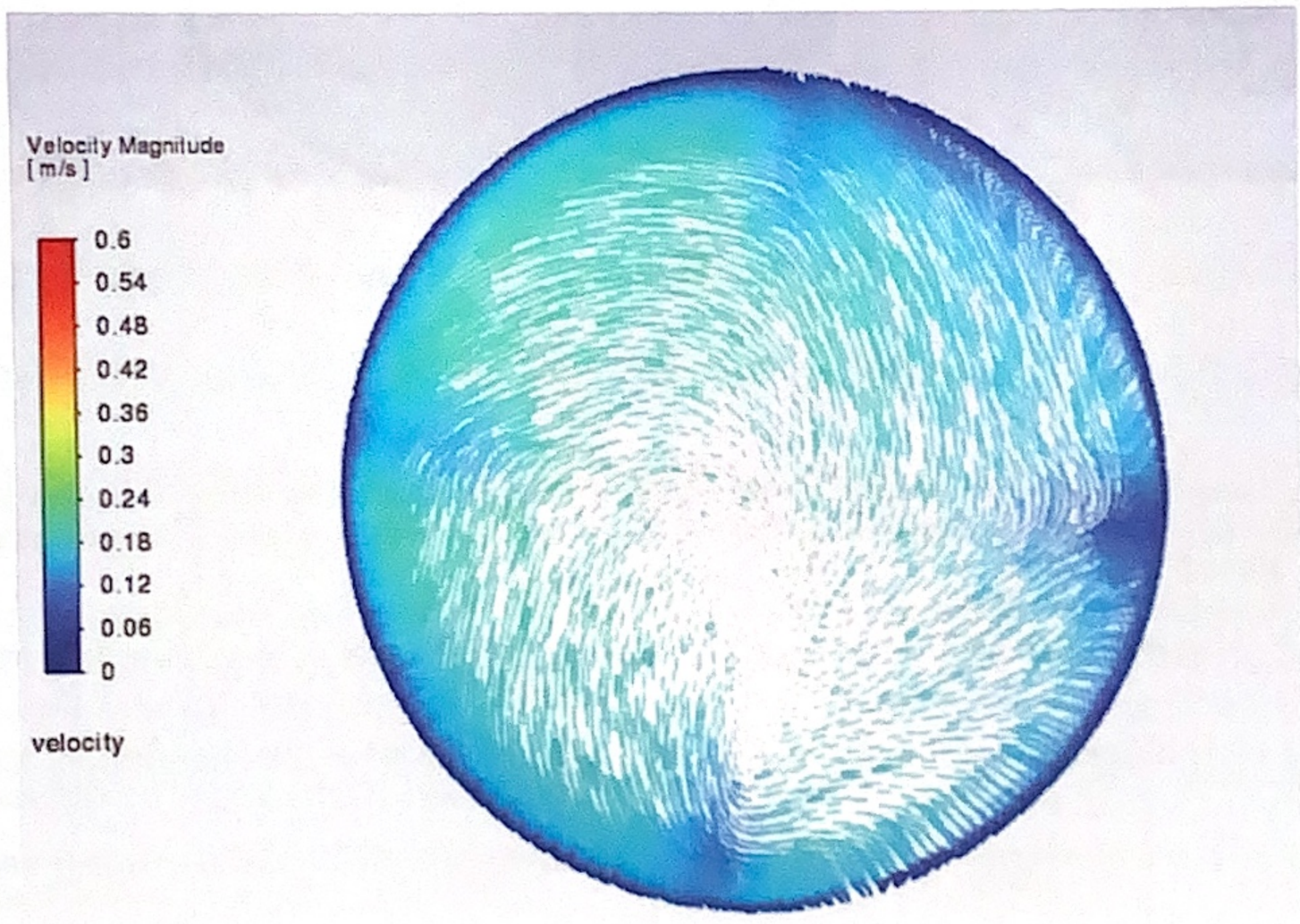


Figure 4-7: Three-dimensional view of the selected vortex-generator configuration (α = 0°, BR = 0.1, PR = 0.2).

The velocity fields shown in Figure 4-8(a) pertains to the values of BR = 0.2, PR = 1.6, and $\alpha = 30$, whereas Figure 4-8(b) refers to the optimal design with BR = 0.1, PR = 0.2, and $\alpha = 0$. A comparison of the two figures clearly reveals the difference in the secondary flow generated from the vortex generator geometry.



(a)

(b)

Figure 4-8: Comparison of cross-sectional velocity fields for non-optimum and optimum vortex-generator configurations: (a) non-optimum configuration ($BR = 0.2$, $PR = 1.6$, $\alpha = 30^{\circ}$), and (b) optimum configuration ($BR = 0.1$, $PR = 0.2$, $\alpha = 0^{\circ}$).

Figure 4-8 (a) shows that the flow is characterized by a high level of asymmetrical rotation, due to the high blockage ratio and the high tilt angle of the winglets that deflect the flow more and create a powerful vortex. This type of flow is beneficial for mixing and may increase heat transfer through disrupting the thermal boundary layer. However, this also implies that there is more flow deflection, greater shear forces, and wake formation downstream. In other words, the physical implications of this include increased form drag and pressure loss.

Figure 4-8 (b) shows the velocity field more uniform and gentler. Despite the presence of the disturbance, there will still be efficiency in mixing because of the flow redistribution in a controlled way. It should be noted that with a smaller blockage ratio, obstruction will not be as severe, and due to zero attack angle, front resistance will be minimized. In other

(b)

Figure 4-8: Comparison of cross-sectional velocity fields for non-optimum and optimum vortex-generator configurations: (a) non-optimum configuration ($BR = 0.2$, $PR = 1.6$, $\alpha = 30^{\circ}$), and (b) optimum configuration ($BR = 0.1$, $PR = 0.2$, $\alpha = 0^{\circ}$).

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words, there will be little fluid separation and pressure loss. Another aspect of this geometry is the low pitch ratio which results in frequent disturbance of the flow through the tube.

4.3 Nanofluids impact

After the vortex-generator optimization established $\alpha = 0°$, BR = 0.1, and PR = 0.2 as the recommended configuration, a second evaluation stage was carried out to improve receiver performance through hybrid nanofluids. The optimized VG geometry was kept fixed while four hybrid nanofluids were screened: Al_2O_3-GO/Syltherm-800, Fe_2O_3-GO/Syltherm-800, Al_2O_3-SiO_2/Syltherm-800, and Cu-MgO/Syltherm-800. The best-performing fluid was then subjected to concentration and mixing-ratio optimization. Table 4-2 shows the thermophysical properties of the base fluid and nanoparticles.

Table 4-2: Thermophysical properties of Syltherm-800 and the nanoparticles used in hybrid nanofluid preparation.

Materials	ρ [kg/m³]	Cp [J/kg·K]	k [W/m·K]	μ [Pa·s]	Reference
Syltherm-800	747.2	1962	0.0961	0.00084	[55]
MgO	3580	874	55	—	[83]
Al_2O_3	3970	765	40	—	[84]
Fe_2O_3	5180	670	6.9	—	[55]
GO	1800	717	5000	—	[85]
SiO_2	2650	730	1.5	—	[86]
Cu	8933	385	401	—	[84]

4.3.1 Screening of Hybrid Nanofluids

The first screening step was performed at a total nanoparticle volume concentration of Φ = 2.0% to compare the candidate hybrid nanofluids on a common basis. Figure 4-9 shows various hybrid nanofluids screening.

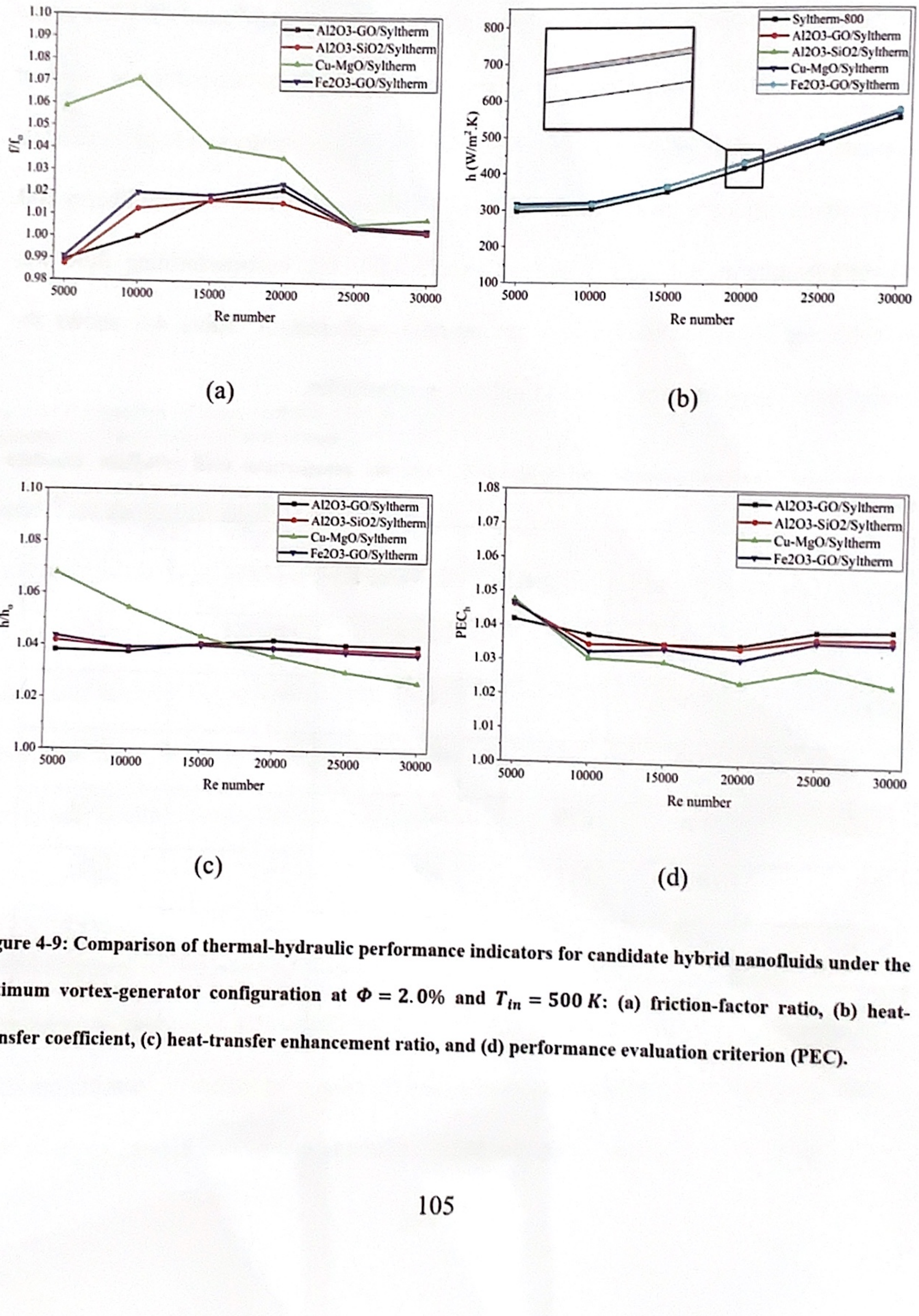


Figure 4-9: Comparison of thermal-hydraulic performance indicators for candidate hybrid nanofluids under the optimum vortex-generator configuration at $\Phi = 2.0\%$ and $T_{in} = 500\,K$: (a) friction-factor ratio, (b) heat-transfer coefficient, (c) heat-transfer enhancement ratio, and (d) performance evaluation criterion (PEC).

It is clear from Figure 4-9 (a) that f/f0 value is mostly equal to one for all cases. However, the nanofluid consisting of Cu-MgO nanoparticles in Syltherm exhibits a high pressure drop factor at low Reynolds number compared to others, but it tends to converge towards other cases by increasing Reynolds number, which demonstrates higher resistance due to viscosity effects. Fe_2O_3-GO/Syltherm and the hybrids made of alumina exhibit negligible variations compared to base fluid.

As can be seen in Figure 4-9 (b), an increase in Reynolds number results in a significant rise in heat transfer coefficient, which is anticipated since the effect of flow enhances the rate of energy transmission. Heat transfer performance in Al_2O_3-GO/Syltherm nanofluid is higher than other cases throughout the whole range and is closely followed by Al_2O_3-SiO_2/Syltherm nanofluid. The lowest enhancements in heat transfer rate are associated with Cu-MgO nanofluid.

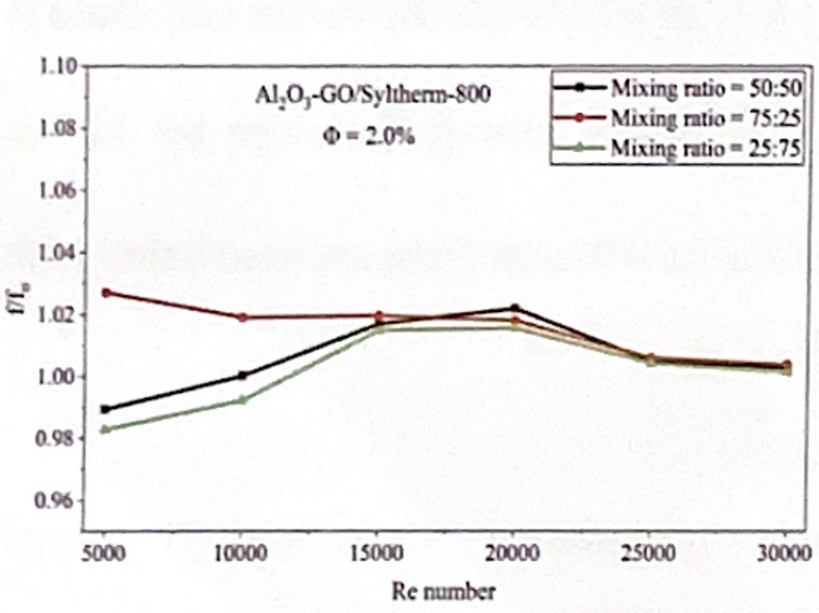
Figure 4-9 (c) shows that Al_2O_3-GO/Syltherm provides a stable and consistent improvement in the heat-transfer coefficient ratio across the investigated Reynolds-number range. Although Cu-MgO/Syltherm-800 also produces considerable enhancement at low Reynolds numbers, its relative advantage decreases as the Reynolds number increases. This occurs because stronger turbulent convection at higher Reynolds numbers reduces the relative contribution of nanoparticle-induced property enhancement.

Lastly, from Figure 4-9 (d) it can be observed that the PEC is used to compare both heat transfer enhancement and pressure loss of the four nanofluids. It is observed that the Al_2O_3-GO/Syltherm nanofluid is the top-performing fluid in terms of the PEC index because it

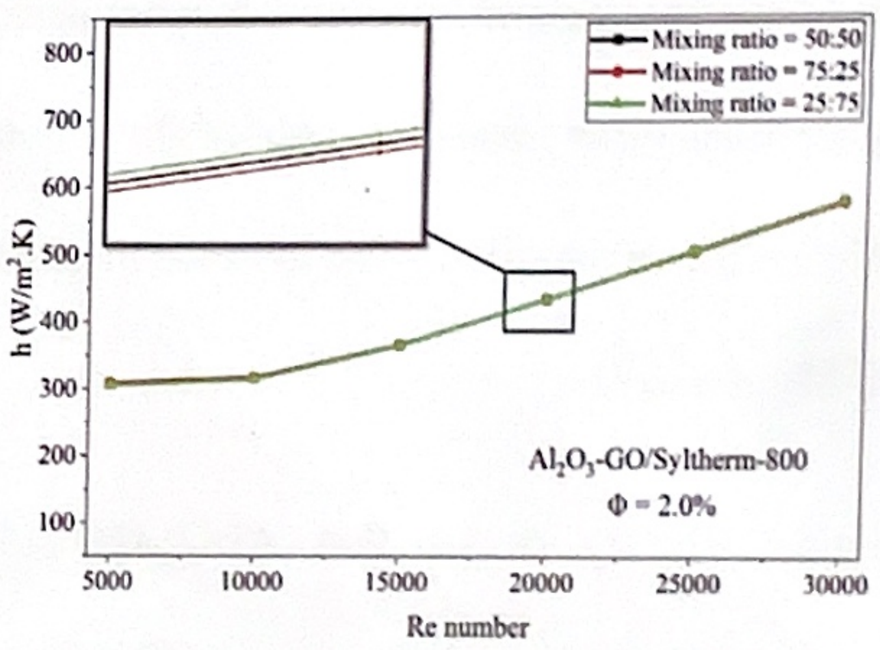
maximizes heat transfer enhancement while keeping hydraulic losses moderate across investigated Re numbers and this due to higher k_{hnf} which improves radial heat diffusion from the absorber wall to the fluid core, increases the convective heat-transfer coefficient. Next comes the Al_2O_3-SiO_2/Syltherm nanofluid while the other two nanofluids perform less effective compared to alumina-based nanofluids.

4.3.2 Effect of Al_2O_3:GO Mixing Ratio

Figure 4-10 shows the second stage of the nanofluid optimization process which examined the Al_2O_3:GO mixing ratio at the selected concentration, considering three compositions: 50:50, 75:25, and 25:75.



(a)



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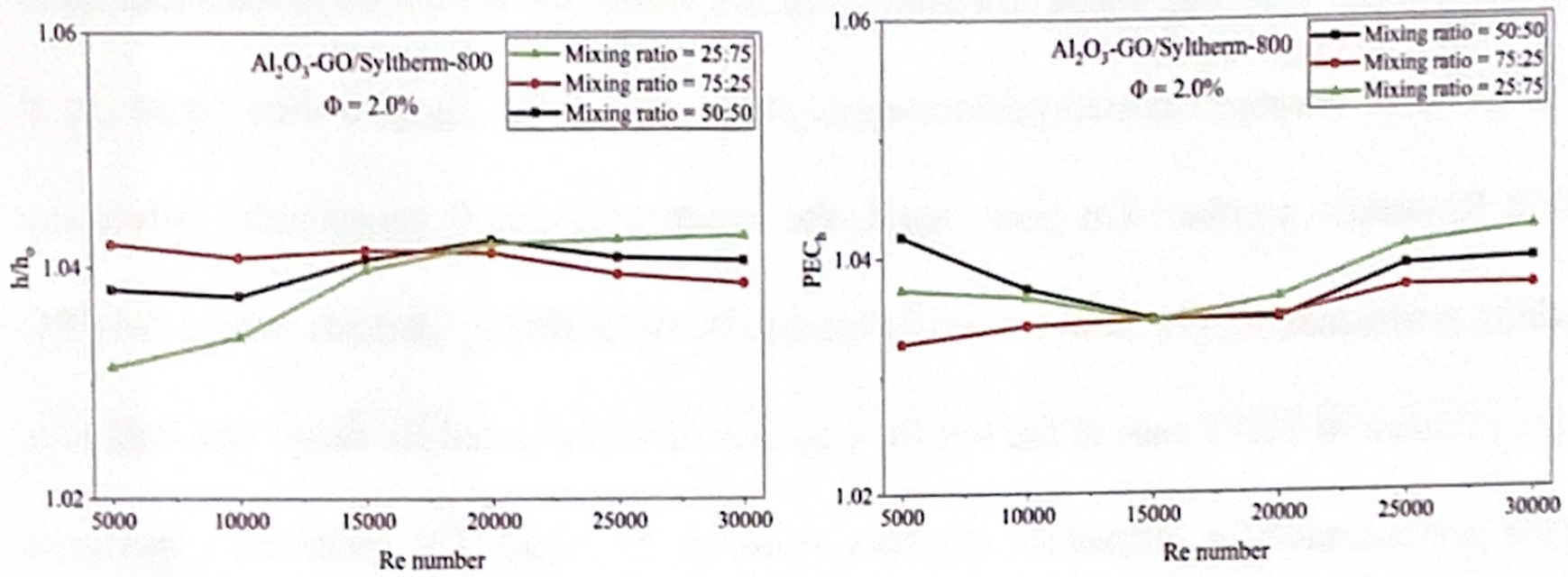


(c) (d)

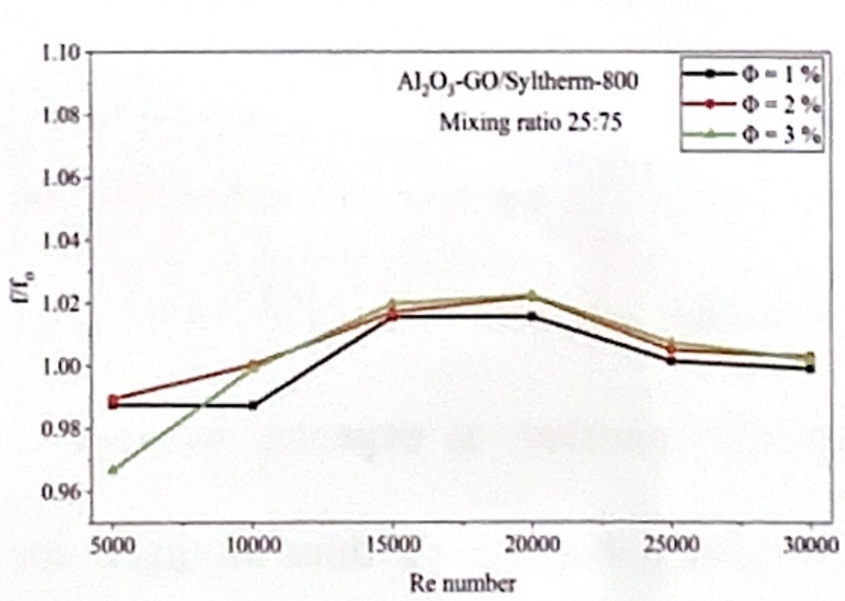
Figure 4-10: Effect of Al_2O_3:GO mixing ratio on thermal-hydraulic performance indicators for Al_2O_3-GO/Syltherm-800 at Φ = 2.0% and Tin = 500 K: (a) friction-factor ratio, (b) heat-transfer coefficient, (c) heat-transfer enhancement ratio, and (d) performance evaluation criterion (PEC).

From Figure 4-10, it is observed that when the mixing ratio of Al_2O_3:GO is varied while maintaining the same concentration level of Φ=2.0%, then the thermal-hydraulic performance is changed in a very subtle way. It can be observed that from Figure 4-10(a), all three cases give f/f0 > 1 from 15000 Re number, meaning that the addition of nanoparticles leads to a higher pressure drop penalty than the base-fluid condition. For all cases, the friction factor is highest for 75:25 and lowest for 25:75, and the 50:50 case lies between them. From Reynolds number variation, it is observed that the difference becomes less significant as Reynolds number increases, implying that the role of mixing ratio on hydraulic resistance is more dominant at lower Reynolds numbers. From Figure 4-10 (b), the heat transfer coefficient increases with Reynolds number, as expected because of increased convective heat transfer. However, the difference between the three mixing ratios is very minimal, suggesting that the change in heat transfer coefficient is not significantly affected by varying Al_2O_3:GO ratios.

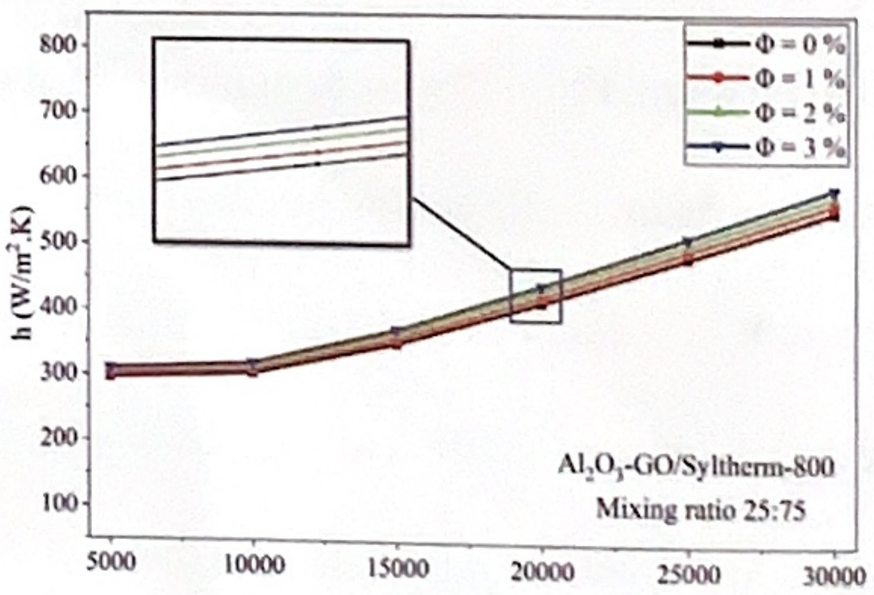
The above point is further made in Figure 4-10 (c), where the ratio h/h0 reveals that the 3 mixtures have varying thermal performance, 25:75 and 75:25 changed their positions at 20000 Reynolds number. On other hand, the mixture of 50:50 consistently offers the medium performance. The above observation can be supported by part (d), where the PEC of the mixture of 25:75 case at the top for majority of investigated Re range and indicates higher performance for further Re numbers, followed by 50:50. The observed behavior is more likely due to the combined effect of the adopted hybrid-property model on effective density, heat capacity, thermal conductivity, and viscosity. Therefore, the 25:75 mixture is selected as the best-performing mixture within the adopted property framework, rather than as a universally optimal physical composition.

4.3.3 Effect of Nanoparticle Volume Concentration for Al_2O_3-GO/Syltherm-800

After selecting Al_2O_3-GO/Syltherm-800 at 25:75 mixing ratio, the total nanoparticle concentration was varied as 1.0%, 2.0%, and 3.0% to determine the most favorable loading level. Concentration of 3% gave the best performance as shown in Figure 4-11.



(a)



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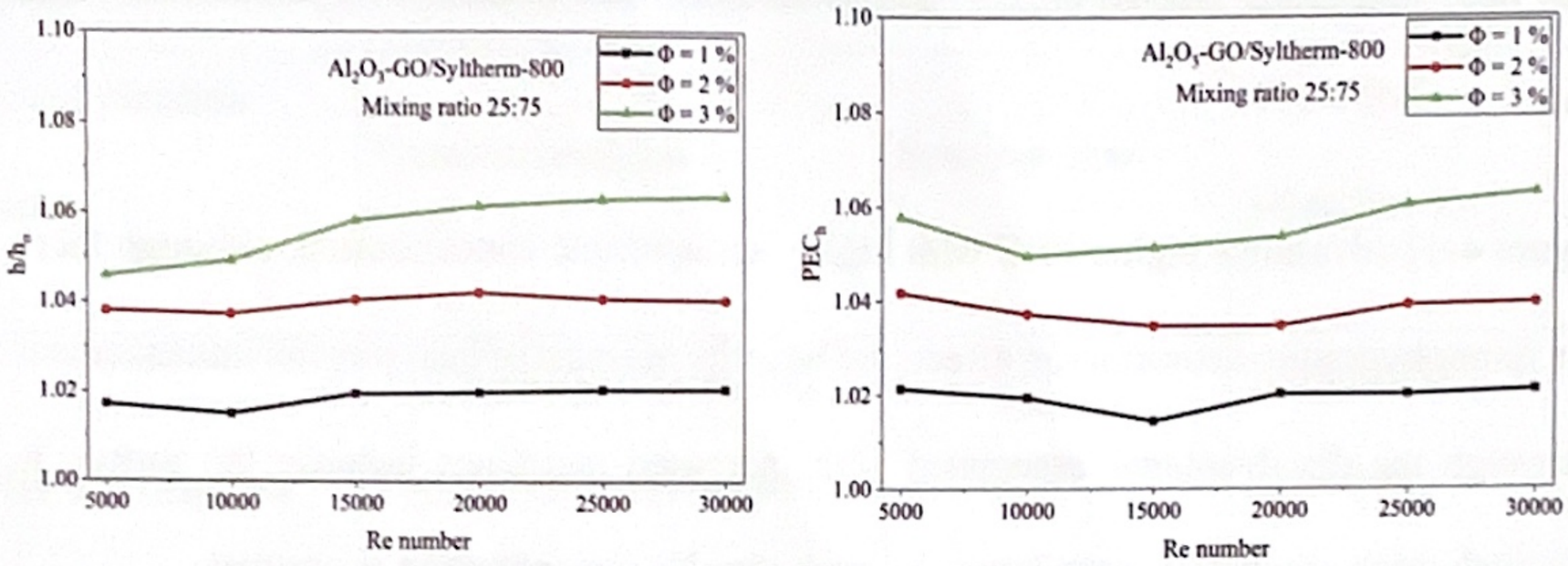


(c) (d)

Figure 4-11: Effect of Al_2O_3-GO/Sylthern nanoparticle volume concentration on thermal-hydraulic performance at $T_{in} = 500\ K$: (a) friction-factor ratio, (b) heat-transfer coefficient, (c) heat-transfer enhancement ratio, and (d) performance evaluation criterion (PEC).

From Figure 4-11 (a), it is evident that the f/f0 values exceed the unity for all concentrations except for low Re numbers and that is due to hotter fluid compared to base fluid which means low viscosity in this low velocities, thus implying that there is some amount of additional pressure drop caused by the addition of nanoparticles. Furthermore, the penalty increases with decrease in Reynolds number and decreases with increase in Re. Additionally, high concentration leads to larger f/f0 because of increased fluid viscosity due to nanoparticle concentration.

Figure 4-11 (b) shows that the heat transfer coefficient increases continually with the Reynolds number. This is caused by the higher convective transport rate at higher Reynolds number. The heat transfer coefficient increases with increase in nanoparticle concentration, and 3% concentration leads to optimal heat transfer coefficient. This occurs because of increased thermal conductivity of the fluid due to the addition of nanoparticles. Similar behavior is observed in Figure 4-11 (c), where the enhancement ratio h/h0 is always greater

than one, suggesting that all concentrations enhance heat transfer compared to the base state.

Figure 4-11 (d) shows higher PEC with higher nanoparticle concentration, although there is a minor hydraulic deterioration effect, the benefits in terms of heat transfer enhancement outweigh the disadvantages associated with increased resistance because the former is relatively high compared to the latter. As such, the 3% concentration is optimal.

4.3.4 Optimum Hybrid Nanofluid and Work Summary

The hybrid-nanofluid screening shows that the best thermal-hydraulic response is not governed by nanoparticle addition alone, but by the combined effect of hybrid type, mixing ratio, and concentration. Among the tested fluids, Al_2O_3-GO/Syltherm-800 gives the highest PEC. Further optimization shows that the 25:75 Al_2O_3-GO mixing ratio and Φ = 3.0% volume concentration provide the most favorable response within the investigated range. Therefore, the final recommended working fluid is Al_2O_3-GO/Syltherm-800 at Φ = 3.0% and Al_2O_3:GO = 25:75. Table 4-3 summarizes all levels of the selection process.

Table 4-3: Summary of hybrid nanofluid screening and selected optimum parameters.

Optimization stage	Cases investigated	Selected case	Basis for selection
Hybrid nanofluid type	Al_2O_3-GO/Syltherm-800; Fe_2O_3-GO/Syltherm-800; Al_2O_3-SiO_2/Syltherm-800; Cu-MgO/Syltherm-800	Al_2O_3-GO/Syltherm-800	Highest PEC across the evaluated nanofluids
Al_2O_3 -GO mixing ratio	50:50, 75:25, 25:75	25:75	Highest PEC
Volume concentration Φ	1.0%, 2.0%, 3.0%	3.0%	Highest PEC within studied range

4.4 Comparative assessment of smooth tube, VG receiver, and VG with hybrid nanofluid receiver

The overall results clearly demonstrate the progressive improvement achieved by moving from the base fluid in a smooth tube to the optimized vortex-generator (VG) configuration, and finally to the hybrid nanofluid combined with VGs. Figure 4-12 shows thermal efficiency, all three cases show a consistent increase with Reynolds number, which reflects the enhancement in convective heat transfer as the flow rate increases. However, the magnitude of improvement differs noticeably among the three configurations. The base

case, consisting of Syltherm-800 flowing through a smooth tube, gives the lowest performance over the entire Reynolds number range. Introducing vortex generators increases both thermal and exergetic efficiencies at all operating conditions, while the combination of hybrid nanofluid and VGs produces the highest values throughout the investigated range. This confirms that the enhancement mechanism is cumulative: the optimized VGs improve the flow structure and thermal mixing, while the hybrid nanofluid further strengthens heat transport because of its superior effective thermophysical properties.

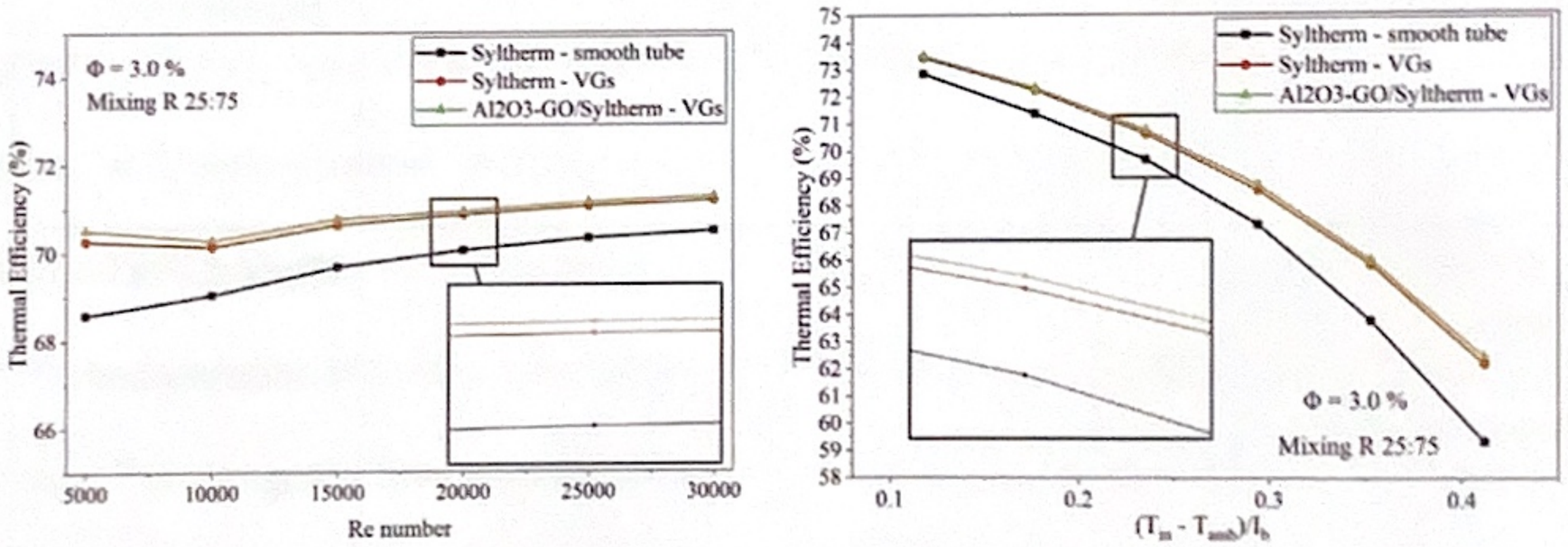


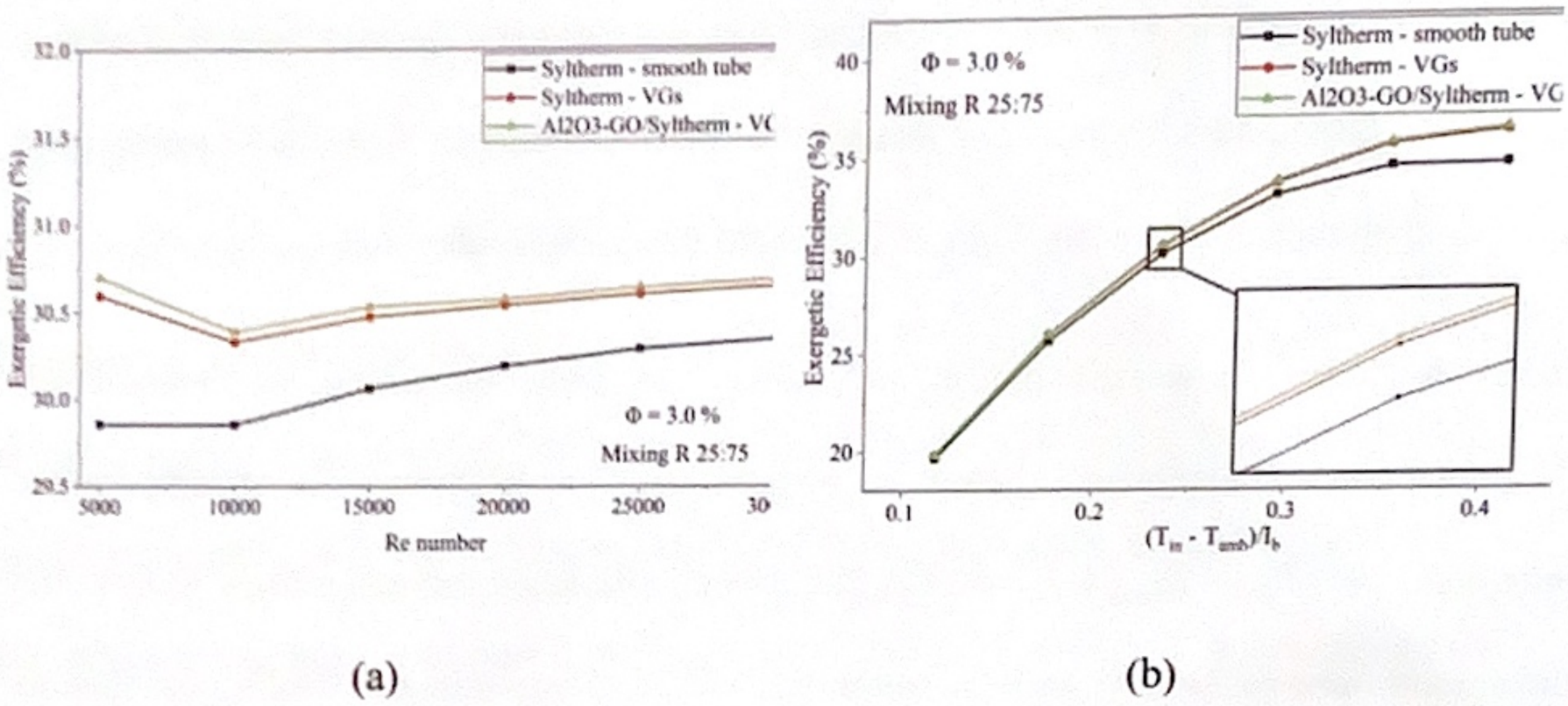
Figure 4-12: Comparison of thermal efficiency for the optimized hybrid nanofluid case, the optimized vortex-generator case, and the smooth-tube base case: (a) thermal efficiency with Re number (b) thermal efficiency with temperature parameter.

For the thermal efficiency change with Re number shown in Figure 16 (a), The enhancement can be separated into two stages: the improvement caused by the optimized VGs and the additional improvement caused by the Al_2O_3-GO/Syltherm-800 hybrid nanofluid. At Re = 5000, the thermal efficiency increased from 68.57% for the smooth tube to 70.26% with the optimized VGs, corresponding to an enhancement of 2.46%. When the

hybrid nanofluid was used with the optimized VGs, the thermal efficiency increased further to 70.51%, giving an additional enhancement of 0.36% compared with the VG-only case and a total enhancement of 2.83% compared with the smooth tube.

Figure 4-12 (b) shows the thermal efficiency variation concerning the temperature parameter the enhancement becomes more pronounced at higher values of temperature parameter. At $(T_{in} - T_{amb})/I_b = 0.12$, the thermal efficiency increased from 72.85% for the smooth tube to 73.45% with the optimized VGs, corresponding to 0.82%, and then to 73.51% with the hybrid nanofluid, giving an additional enhancement of 0.07% and a total enhancement of 0.90%.

At temperature parameter = 0.41, the thermal efficiency increased from 59.33% for the smooth tube to 62.17% with the optimized VGs, corresponding to 4.79%, and then to 62.42% with the hybrid nanofluid, giving an additional enhancement of 0.40% and a total enhancement of 5.21%.



(a) (b)

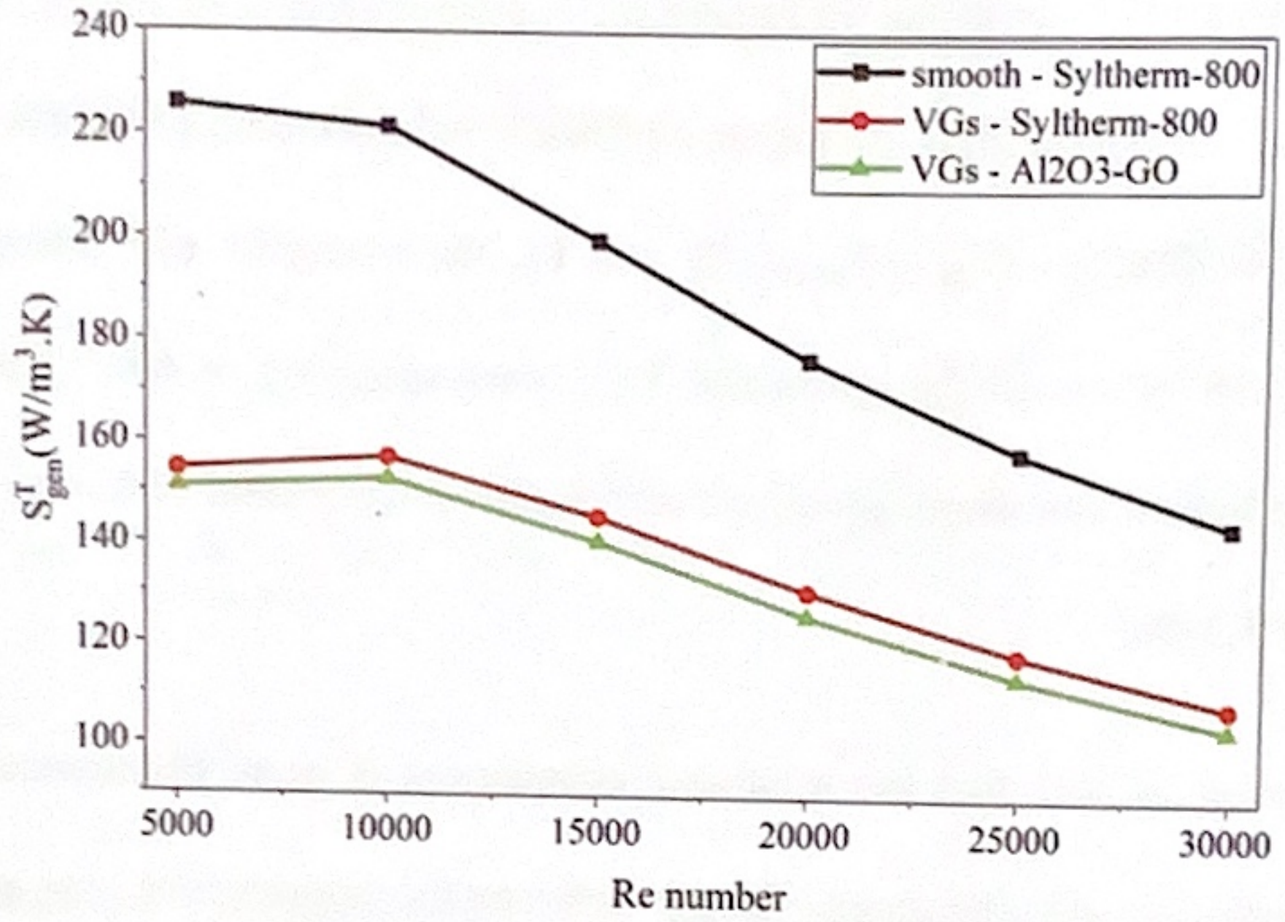
Figure 4-13: Comparison of exergetic efficiency for the optimized hybrid nanofluid case, the optimized vortex-generator case, and the smooth-tube base case: (a) exergetic efficiency with Re number (b) exergetic efficiency with temperature parameter.

Unlike thermal efficiency, exergetic efficiency evaluates the useful work potential of the recovered heat. Figure 4-13 (a) shows that, the exergetic efficiency followed the same cumulative enhancement trend. At Re = 5000, the exergetic efficiency increased from 29.86% for the smooth tube to 30.60% with the optimized VGs, corresponding to an enhancement of 2.48%. With the addition of Al_2O_3-GO/Syltherm-800, the exergetic efficiency increased to 30.70%, giving an additional enhancement of 0.34% compared with the VG-only case and a total enhancement of 2.83% compared with the smooth tube. At Re = 30000, the exergetic efficiency increased from 30.34% to 30.65% with the optimized VGs, giving an enhancement of 1.00%, and then to 30.69% with the hybrid nanofluid, corresponding to an additional enhancement of 0.13% and a total enhancement of 1.13%.

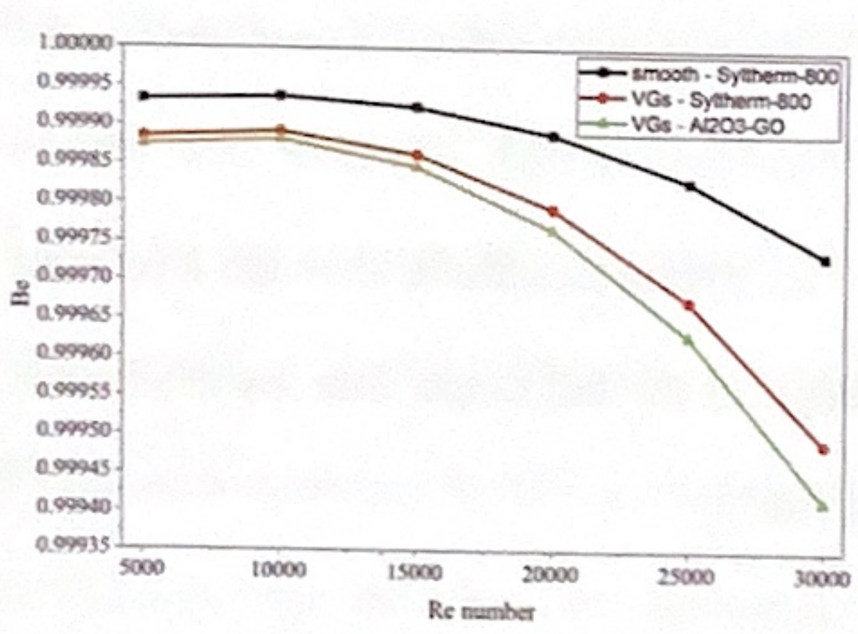
Figure 4-13 (b) shows the variation of the exergetic efficiency with the temperature parameter, the exergetic-efficiency enhancement also increased at higher temperature parameter. At $(T_{in} - T_{amb})/I_b = 0.12$, the exergetic efficiency increased from 19.66% for

the smooth tube to 19.82% with the optimized VGs, corresponding to 0.83%, and then to 19.83% with the hybrid nanofluid, giving an additional enhancement of 0.06% and a total enhancement of 0.90%. At $(T_{in} - T_{amb})/I_b = 0.41$, the exergetic efficiency increased from 34.48% to 36.14% with the optimized VGs, corresponding to 4.81%, and then to 36.29% with the hybrid nanofluid, giving an additional enhancement of 0.40% and a total enhancement of 5.23%.

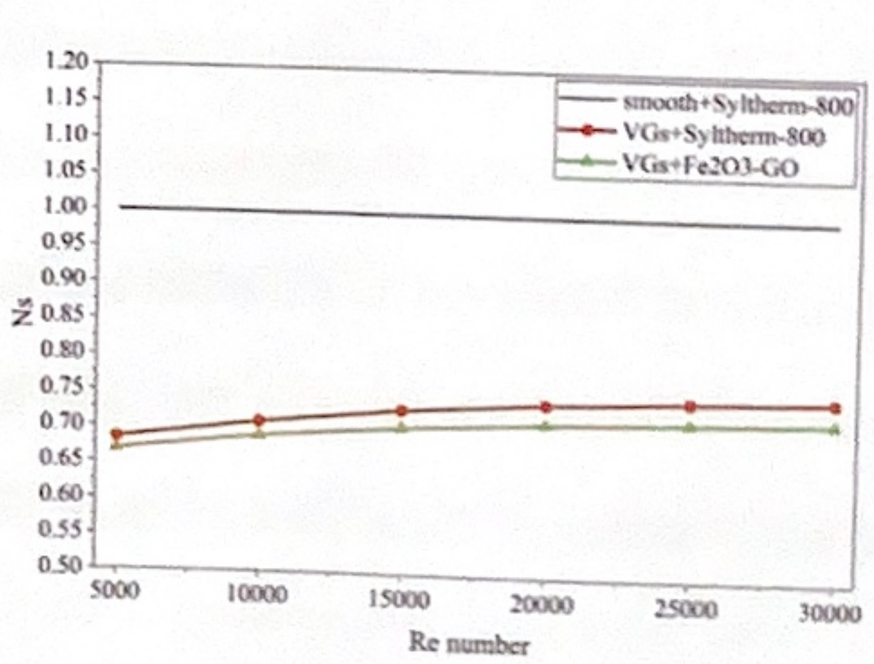
It is also important to note that the relative enhancement is more pronounced at lower Reynolds numbers than at higher ones. The gap between the smooth-tube case and the VG-assisted cases is largest in the low-Reynolds-number region, whereas the difference gradually narrows as Reynolds number increases. This behavior is expected because, at higher Reynolds numbers, the base flow already has stronger inertial and turbulent effects, which naturally improve convective heat transfer even without inserts. Consequently, the additional contribution of VGs becomes less dominant at high Reynolds numbers. A similar trend is observed for the added benefit of the hybrid nanofluid, although it remains consistently superior to the other two cases throughout the full range. Entropy analysis in Figure 18 confirmed thermal and exergetic behavior.



(a)



(b)



(c)

Figure 4-14: Comparison of entropy generation versus Reynolds number for the smooth tube, optimized vortex-generator, and optimized hybrid nanofluid cases. (a) Total entropy generation. (b) Bejan number. (c) Entropy generation ratio (Ns).

The total entropy generation findings serve as another validation of the thermal and exergetic behavior. As illustrated in Figure 4-14 (a), the entropy generation drops with

increasing Reynolds numbers in all three configurations, suggesting less irreversibility due to increased convective flow leading to homogeneity in the temperature distribution within the receiver. The pure Syltherm smooth tube configuration exhibits the highest value of entropy generation across the whole range of Reynolds numbers since the enhanced thermal irreversibility can be attributed to the lesser fluid mixing ability and poor heat transfer characteristics. The addition of vortex generators results in a lower level of entropy generation since the formation of secondary flow leads to higher mixing, thinner thermal boundary layer, and lower irreversibility caused by the temperature gradient. The Al_2O_3-GO/Syltherm nanofluid configuration with vortex generators gives the lowest entropy generation, proving that this configuration has the best heat transfer efficiency and optimal thermodynamic energy usage. Thus, the entropy generation findings confirmed the findings on thermal efficiency and exergetic efficiency.

The Bejan number results in Figure 4-14(b) remain very close to unity for all cases, showing that entropy generation is governed primarily by heat-transfer irreversibility rather than fluid-friction irreversibility. This observation is important because it confirms that the main thermodynamic losses in the present PTSC receiver are associated with temperature gradients, not viscous dissipation. Figure 18 (c) supports this interpretation, where the entropy-generation ratio remains below unity and shows only limited variation with Reynolds number. Thus, even though vortex generators increase pressure drop, the resulting frictional irreversibility is still much smaller than the thermal irreversibility, and the net effect remains favorable.

In general, these findings show that the improved design of delta-winglet effectively increases the performance of the receiver mainly through better radial mixing and the

repeated disturbance of the thermal boundary layer. The final combination, with $\alpha = 0°$, BR = 0.1, and PR = 0.2, offers an appropriate balance between thermal performance and pressure drop, since it offers significant enhancement in heat transfer without increasing the pressure drop too much. Inclusion of the optimal nanofluid also helps improve thermal performance and exergy efficiency owing to its increased effectiveness in thermal energy transport of the fluid. Figure 4-17 shows the local entropy-generation contours.

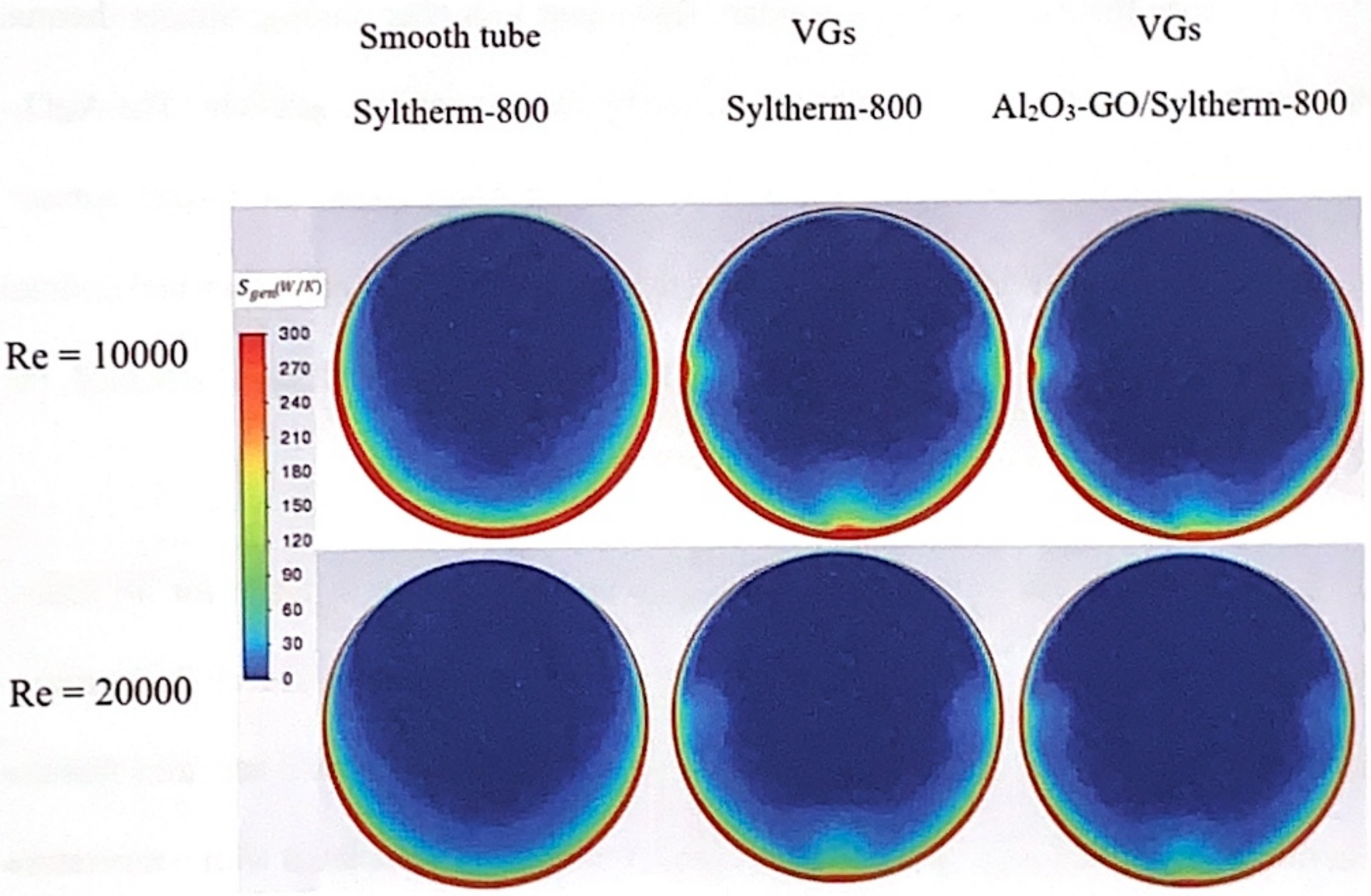


Figure 4-15: Local entropy generated at section of absorber tube at 10000, 20000 Re number and inlet temperature 500 K.

In all cases, the highest local entropy generation is concentrated near the tube wall, while the core region remains at much lower values. The smooth-tube case exhibits the widest and strongest high-entropy region, indicating the highest level of irreversibility. After inserting the optimized vortex generators, the entropy-generation region becomes weaker and less extensive, which indicates a reduction in irreversibility due to improved mixing

and more effective heat transport. A further reduction is observed when Al_2O_3-GO/Syltherm is used with the optimized VGs, confirming that the hybrid nanofluid enhances the heat-transfer process and reduces local irreversibility more effectively than Syltherm-800 alone. In addition, increasing Reynolds number from 10000 to 20000 decreases the local entropy-generation level for all cases, showing that stronger convection reduces the overall irreversibility inside the receiver. Therefore, the contour plots confirm that the combined use of optimized VGs and Al_2O_3-GO/Syltherm provides the most thermodynamically favorable configuration among the studied cases.

Overall, the results establish a clear performance ranking: optimized vortex-generator receiver with Al_2O_3-GO/Syltherm, optimized vortex-generator receiver with Syltherm-800, and smooth tube with Syltherm-800. The smooth tube represents the baseline case and gives the lowest thermal and exergetic efficiencies. The optimized vortex generators improve performance by generating secondary flow and disturbing the thermal boundary layer while maintaining an acceptable hydraulic penalty. The best-performing configuration is obtained when the optimized vortex generators are combined with Al_2O_3-GO/Syltherm-800, where geometric enhancement and improved fluid thermal properties act together to produce the highest thermal and exergetic performance.

4.5 Proposed Correlations

The proposed correlations were developed to provide direct predictive equations for the heat transfer coefficient, friction factor, and thermal efficiency of the parabolic trough receiver with vortex generators and Al_2O_3-GO/Syltherm-800 under different operating conditions. The correlations were obtained by fitting the CFD results using a nonlinear least-squares regression method, where the model constants and exponents were adjusted

to minimize the total sum of squared errors between the predicted values and the numerical data. Therefore, the proposed equations represent compact semi-empirical models that are valid within the investigated operating range.

$$h = 6.0568 \times 10^{-7} Re^{1.9324} + 0.07118 \left(\frac{Pr}{1-\phi}\right)^{2.0891} + \frac{10023.79}{T_{in}^{0.5872}} + 0.04739 Re\phi \quad (4.1)$$

$$f = 0.02105 \left(\frac{Pr}{1-\phi}\right)^{0.2209} + 2.9682 \times 10^{-15} \frac{T_{in}^{8.0063}}{Re^{2.2206}} \quad (4.2)$$

$$\eta_{th} = 73.5198 + 2.8236 \times 10^{-5} Re^{1.0307} - 81.2378 \left(\frac{T_{in} - 300}{850}\right)^{2.1777} + 4.6226\phi \quad (4.3)$$

$$5000 \leq Re \leq 30000$$
$$400 \leq T_{in} \leq 650 \text{ K}$$
$$0 \leq \phi \leq 0.03$$

The accuracy of the proposed correlations confirms their strong alignment with the CFD results. The heat transfer coefficient correlation achieved $R^2 = 0.9942$, with maximum error of 4.22%. The friction factor correlation showed $R^2 = 0.9953$, with maximum error of 5.34%. Similarly, the thermal efficiency correlation provided the highest agreement, with $R^2 = 0.9972$ and maximum terror of 0.53%. These high R^2 values and low maximum errors indicate that the proposed correlations can reliably estimate the thermo-hydraulic performance of the receiver without repeating the full CFD simulations. This is particularly useful for parabolic collector analysis, where such correlations can support rapid performance prediction, parametric studies, optimization, and preliminary design of receivers operating with hybrid nanofluids and vortex-generator inserts.